Agrivive: QR-Based Surplus Wet Marketplace with Automated Promotion and Contextual Seller Intelligence

Arianne B. Bartiquien
BS in Information Technology
University of Mindanao
a.bartiquien.546699@umindanao.edu.ph

Jara Joaquin T. Jara
BS in Information Technology
University of Mindanao
j.jara.546428@umindanao.edu.ph

Paula Shayne B. Marundan
BS in Information Technology
University of Mindanao
p.marundan.546642@umindanao.edu.ph

Charisse P. Barbosa
University of Mindanao
charisbarbosa@umindanao.edu.ph

CCS Concepts
CCS → Information systems → Information systems applications → Decision support systems→ Data Analytics

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1. INTRODUCTION

1.1 Project Context

Digital marketplace platforms connect sellers and buyers through structured listings, communication tools, and transaction-support services [1]. Product photographs, descriptions, prices, quantities, seller profiles, and pickup details help reduce information gaps, while combined physical and digital selling channels can broaden market reach and improve access to agricultural products [2].

At the global level, food loss and waste generate economic, environmental, and social costs because the resources used in food production and distribution are also wasted [3], [4]. Approximately 1.05 billion tonnes of food waste were generated worldwide in 2022 [3], while selected treatment methods may cost about US$138 per tonne [4].

In the Philippines, postharvest losses exceeding 20% have been reported for cabbage, eggplant, and tomato because of overproduction, improper handling, limited storage, transportation problems, and inadequate monitoring [5], [6]. Tomato losses also reached 10.94% along the Nueva Ecija–Manila chain and 24.14% along the Bukidnon–Manila chain, supporting recommendations for online trading and digital marketing [7], [8]. Although Republic Act No. 9003 addresses ecological solid-waste management [9], Republic Act No. 9803 encourages food donation [10], and Republic Act No. 7581 provides the national framework for price monitoring and controls [11], these measures do not establish a localized mechanism for tracking and marketing vegetables while they remain marketable surplus. Moreover, although digital agriculture can improve information exchange, operational efficiency, and market access, its adoption remains constrained by connectivity, affordability, digital literacy, device access, and the capacity of small agricultural enterprises [12], [13].

Within this study, surplus vegetables refer to seller-declared, marketable vegetables intended for human consumption that exceed expected or confirmed buyer demand and remain unsold within their intended selling period [14]. In Agrivive, the term sellers collectively refers to vegetable suppliers, supplier-vendors, and retail vendors who post marketable surplus vegetables through the platform. The local scope covers sellers and buyers in Davao City. Data were gathered and validated at Bankerohan Public Market, Agdao Public Market, and the Davao Food Terminal Complex, with consultations involving the City Agriculturist's Office and City Economic Enterprises Office. Respondents included 38 produce sellers and 41 buyers from households and food-related businesses. Some participants performed overlapping roles, such as supplying regular or bulk buyers while also selling to walk-in customers or sourcing vegetables for resale.

In practice, sellers often encounter situations where the quantity of vegetables prepared or acquired exceeds actual demand, existing studies show that surplus occurs when the quantity prepared or acquired exceeds actual demand. Overstock may result from reduced or cancelled orders, changes in regular-buyer demand, simultaneous supply, seasonal conditions, holidays, school breaks, weather changes, low customer traffic, temporary closures, and price competition [14]. Fresh-produce purchases also vary according to seasonal, economic, product-related, and situational factors [15], making demand difficult to know when sellers rely on previous transactions, anticipated sales, or regular orders.

These findings were validated by the surveyed and interviewed sellers. Surplus occurred when regular or bulk buyers reduced, cancelled, or transferred orders because of fewer customers, menu changes, limited budgets, remaining inventory, holidays,

weather, or temporary closures. Sellers likewise experienced surplus when the quantity purchased for resale exceeded walk-in demand. For example, a supplier preparing 500 kilograms of cabbage but selling only 300 kilograms retains 200 kilograms of surplus, while a vendor purchasing 30 kilograms and selling 20 kilograms retains 10 kilograms, provided that the vegetables remain marketable.

Research further shows that agricultural sellers use physical and digital channels to reach customers [12]. In practice, sellers primarily depend on physical stalls, walk-in customers, regular buyers, personal contacts, direct offers, phone calls, text messages, Messenger, and Facebook groups when attempting to sell their remaining vegetables. They first attempt to dispose of surplus by posting available vegetables in Facebook groups and other social media platforms. Based on the interviews, both suppliers and vendors then reduce prices on these surplus items to attract buyers and recover part of their capital. Suppliers may also create their own posts on Facebook to further promote remaining stocks when initial efforts do not result in sufficient sales.

Nevertheless, visibility depended on group membership, timing, and platform algorithms, preventing sellers from consistently reaching broader buyers. Given the limited marketable life of vegetables, sellers also faced increasing pressure to sell before product quality declined. All surveyed sellers lowered prices on unsold but marketable vegetables, 94.7% were willing to sell at or near buying price to recover capital, and 97.4% reported that unsold and spoiled vegetables reduced their weekly take-home income. Thus, unpredictable fluctuations in buyer demand, limited market reach, fragmented communication, and short product life increase the risk of incomplete capital recovery and income loss.

Similarly, previous research indicates that buyers vary their purchasing quantity, frequency, supplier choice, and transaction method according to customer demand, menu requirements, inventory, budget, price, product condition, supplier reliability, storage capacity, perishability, holidays, school breaks, special events, weather, and temporary closures [16]. Local surveys and interviews confirmed that buyers' decisions depended on immediate household or business needs, remaining stock, expected customers, available budget, and suitability for cooking. Although all 41 buyers were willing to purchase excess or visually unattractive vegetables when suitable for cooking and fairly priced, they could remain unaware of available supplies when information was scattered across stalls, personal networks, Facebook groups, and separate social-media posts.

Furthermore, digital-marketplace studies show that photographs, descriptions, prices, quantities, seller profiles, and pickup information are factors considered by buyers when evaluating products and sellers [17]. Consistent with this, 95.1% of local buyers wanted recent product photographs before travelling, while 97.6% preferred evidence of seller legitimacy. All buyers were willing to use self-pickup to avoid delivery fees, making distance, exact location, and pickup instructions important. These conditions indicate that changing purchasing requirements, incomplete product information, inconsistent social-media visibility, location uncertainty, and limited seller verification are associated with difficulties in product discovery, trust, and transaction coordination.

At the local level, existing laws and government activities address waste management, food donation, agricultural marketing, and commodity-price monitoring. However, these efforts do not specifically document or monitor marketable surplus vegetables and their transactions within Davao City. This limitation was observed across the three data-gathering sites, where transactions remained fragmented among physical markets, regular-buyer networks, personal contacts, calls, messages, Messenger, and Facebook groups. As a result, information on available products, quantities, conditions, seller locations, and changing buyer needs remained dispersed, contributing to mismatches between sellers with surplus and buyers seeking corresponding supplies.

Furthermore, consultations with the City Agriculturist's Office and City Economic Enterprises Office indicated that local records primarily focus on administrative operations and commodity prices. There are no consolidated records capturing expected and remaining supplies, surplus quantities, quantities offered and sold, recurring overstock, completed transactions, changes in buyer demand, or seasonal marketplace activity. Consequently, sellers, market administrators, and stakeholders have limited historical information for determining how often surplus occurs, which vegetables commonly remain unsold, and how demand changes over time.

In comparison, existing systems demonstrate different methods of improving digital selling and agricultural distribution. Facebook Marketplace uses seller-created listings, photographs, locations, and Messenger communication for general local trade [18]. This is relevant because it demonstrates digital product visibility and direct communication; however, it does not distinguish marketable vegetable surplus, bulk and smaller quantities, or recurring overstock.

Similarly, eKadiwa uses government-supported web ordering to connect agricultural producers and consumers [19]. It demonstrates how an online platform can expand access to farm products, but it focuses on general agricultural selling and program-based participation rather than independent, time-sensitive surplus listings. DELIVER-e integrates e-commerce, blockchain-supported supply-chain processes, warehousing, and logistics coordination [20]. Although relevant for structured agricultural distribution, it does not address daily surplus from individual sellers.

Meanwhile, Ruri Club organizes Rescue Buy and Group Buy campaigns for excess produce [21]. Its campaign model demonstrates how surplus may be marketed collectively, but listings are organization-managed and do not provide individual sellers with records of recurring surplus. Lastly, MooMart connects farmers with food and beverage businesses through a business-to-business marketplace [22]. While it demonstrates direct agricultural market linkage, it focuses on regular sourcing rather than smaller seller surplus, changing daily conditions, and localized Davao City records.

Table 1. Comparison Matrix of Existing Agricultural and Marketplace Systems

Evaluation criterion	Facebook Marketplace	eKadiwa	DELIVER-e	Ruri Club	MooMart	Agrivive
User navigation and interface	Familiar interface, but users browse a broad range of products, which may require additional searching to locate relevant surplus	Structured agricultural ordering, but users may need to navigate program-specific product and ordering functions	Multiple supply-chain functions may require users to process different ordering and distribution information	Campaign-oriented interface focused on surplus campaigns rather than individual surplus transactions	B2B-oriented interface suited to business sourcing and larger transactions	Task-focused navigation with filtering, clear information, reservation, QR pickup, and location features to reduce steps and cognitive load.
Search and multi-criteria filtering	Basic search and browsing	Product/order browsing	Supply-chain search	Campaign browsing	Product sourcing	Multi-criteria filtering using product/category, price, location, and availability to help buyers quickly find suitable surplus
Surplus-specific listing	Not designed specifically for surplus	General agricultural products	General distribution	Surplus campaigns	Regular farm products	Dedicated marketable-surplus listings with quantity, price, condition, and surplus information
Product condition and quantity visibility	Basic photos and descriptions	General product details	Supply and order details	Campaign details	Product/source details	Displays current product information, condition, available quantity, pricing, and listing details before reservation
Seller geolocation	Location may be provided by seller	Program/location information	Supply-chain locations	Organization/campaign location	Seller/source information	Seller geolocation and pickup instructions through map-based location information.
Reservation / booking mechanism	Buyer-seller direct coordination	Online ordering	Integrated ordering and logistics	Campaign-based ordering	Business-order processing	Reservation process that changes available quantity to reserved quantity and records the transaction
QR-based pickup verification	Not indicated	Not indicated	Not indicated	Not indicated	Not indicated	Generates a unique QR code for reservations and uses QR scanning to verify pickup and handover.
Automated promotion	Manual posting	Government promotion partnerships	Partnerships and logistics networks	Campaign/social promotion	Business networking and partnerships	Automated promotional workflow generates promotional content, sends buyer notifications, and supports social-media sharing
Time-sensitive listing management	General listing management	Program-managed listings	Supply-chain management	Campaign management	Marketplace management	Tracks active listings and reservation expiration, restores quantities from expired/cancelled reservations, and updates listing status automatically
Transaction and surplus records	No dedicated surplus records	No seller-level surplus history	No localized surplus records	No individual surplus tracking	No surplus records	Records posted, reserved, sold, completed, and cancelled transactions for seller-level surplus monitoring
Seller-level descriptive analytics	Limited seller information	Program-managed information	Supply-chain records	Campaign information	Business information	Provides descriptive sales and surplus trend information based on recorded marketplace transactions
Verification and accountability	Profiles, ratings, reports	Program-managed participation	Partner-managed participation	Organization-managed participation	Seller information	Verified seller accounts and transaction-linked records provide stronger accountability for marketplace interactions
User communication	Messaging	Program/order communication	Integrated logistics communication	Campaign communication	Business communication	In-app messaging and notifications support coordination between buyers and sellers

As shown in Table 1, existing systems provide marketplace, ordering, logistics, and promotional functions, but they do not combine the technical features needed to address the specific problems of localized surplus vegetable transactions. Agrivive uses task-focused navigation to reduce unnecessary steps when users search, evaluate, reserve, and collect products, helping address difficulty in navigating scattered information. Multi-criteria filtering addresses the problem of buyers having to browse many listings by allowing them to narrow products based on category, price, location, and availability. Clear information on product condition, quantity, and price reduces uncertainty and limits the need for repeated messages before making a purchase. The reservation feature addresses unclear booking and stock availability by separating available and reserved quantities, while the QR code provides a unique reservation reference that helps verify the buyer and confirm the correct pickup, reducing confusion during handover using Next.JS. Seller geolocation and pickup instructions address difficulty in locating sellers by giving buyers a clear pickup point and directions using OpenStreetMap and React Leaflet. Automated promotion, notifications, and social-media sharing address limited surplus visibility and repeated manual posting by giving sellers additional ways to reach potential buyers using n8n. Time-sensitive listing management addresses outdated availability by automatically updating listing status and restoring quantities from expired or cancelled reservations. Transaction records and descriptive analytics address the lack of organized historical information by allowing sellers to review posted, reserved, sold, and recurring surplus activity. Thus, Agrivive combines these technical approaches to directly address problems in product discovery, information access, reservation, pickup coordination, promotion, and surplus monitoring.

1.2 Purpose and Description

This study aims to develop Agrivive: QR-Based Surplus Wet Marketplace with Automated Promotion and Contextual Seller Intelligence, a cross-platform marketplace information system for vegetable suppliers, supplier-vendors, vendors, and buyers in Davao City. The project aligns with the University of Mindanao's research agenda on Digital Transformation and Sustainable Community Development by applying digital marketplace technology, workflow automation, and data analytics to a local agricultural concern. It supports SDG 2 – Zero Hunger by improving access to marketable vegetables, SDG 8 – Decent Work and Economic Growth by creating additional selling opportunities for Sellers, and SDG 12 – Responsible Consumption and Production by supporting the redistribution of surplus vegetables before their marketability declines.

Agrivive consists of a mobile application for verified sellers and a web application for buyers, and the public. Its primary functions include structured bulk and smaller-quantity surplus listings, AHP-TOPSIS-based multi-criteria weighted listing visibility , automated promotional-content preparation through n8n, seller-declared product information, seller-location and pickup access, buyer–seller communication, 24-hour QR-based reservations, pickup confirmation, descriptive surplus analytics, calendar advisories, and transaction-linked trust controls. Sellers may manage listings and prices, review visibility levels and promotional content, coordinate reservations, scan QR codes during pickup, and view historical marketplace summaries. Buyers may search and filter listings, evaluate product and seller information, communicate with sellers, reserve available quantities, receive time-limited QR codes, locate pickup points, and report transaction related and communication concerns. Seller verification and user and listing management are handled through system processes, while a rule-based flagging mechanism uses a weighted statistical tool to evaluate reported activity and determine corresponding account actions based on configured thresholds. The public may view only aggregated and non-personally identifiable marketplace information for monitoring and planning. Agrivive provides decision-support information based on recorded system data and does not guarantee sales, certify food safety or internal freshness, or predict future demand.

1.3 Objectives

1.3.1 General Objectives

This study aims to develop Agrivive, a cross-platform surplus marketplace information system built on Next.JS, React Native, ElysiaJS, and PostgreSQL, enabling vegetable suppliers, vendors, and buyers in Davao City to register, verify roles, list structured surplus inventory, and coordinate reservation and pickup coordination with transparent transaction monitoring and stakeholder accountability.

1.3.2 Specific Objectives

To attain the general objective, the researchers develop a system to:

1.3.2.1 Implement a visibility tier system (Priority Boost, Standard Exposure, and Basic/Hold) that ranks active listings and routes qualified ones to n8n for automated promotional content and buyer alerts using AHP–TOPSIS Multi-Criteria Decision Visibility Engine.

1.3.2.2 Generate QR-code-based reservation using Next.JS React QR and self-pickup coordination with 24-hour validity, real-time quantity tracking (available/reserved/sold), and seller location display using React Leaflet.

1.3.2.3 Generate analytics dashboards and contextual advisories showing sell-through rates, frequently posted surplus vegetables, completed transactions using Descriptive Sales Trend Analytics, and calendar-based planning reminders using the Nager.Date API.

1.3.2.4 Create a rating and review mechanism that uses a weighted statistical tool to evaluate transaction ratings and reported user activity, allowing users to rate transactions, provide reviews, and report suspected fraud for transaction monitoring and accountability.

1.4 Scope and Limitation

This study covers the design, development, and beta testing of Agrivive, a cross-platform marketplace information system for marketable surplus vegetable transactions during Academic Year 2026–2027. The system includes a React Native mobile application for verified vegetable suppliers, supplier-vendors, and retail vendors, and a Next.JS web application for household and food-business buyers, administrators, the public, and stakeholders from the City Agriculturist's Office, City Economic Enterprises Office, and other offices. It is supported by ElysiaJS, PostgreSQL, Better Auth, Hostinger, n8n for marketing automation and Python for system processing. Agrivive features account registration and seller verification; bulk and small-quantity surplus listings; product search and multi-criteria filtering; recent photographs and seller-declared condition details; buyer-seller

communication; notifications; shareable promotional content; seller geolocation with pickup instructions; reservation and self-pickup confirmation; 24-hour QR-based reservation; and real-time quantity tracking for available, reserved, and sold products. The system also provides sellers with descriptive analytics on posted, reserved, sold, and unsold surplus, contextual holiday advisories. Ratings, reviews, and fraud reports use a weighted statistical tool. The public and stakeholders may access aggregated and non-personally identifiable marketplace information. The study focuses on users within Davao City, with data gathering and beta testing conducted among selected participants from Bankerohan Public Market, Agdao Public Market, and the Davao Food Terminal Complex.

The study excludes meat, poultry, seafood, processed food, and non-agricultural products due to differing handling and regulatory requirements. It does not process online payments, verify transactions, issue refunds, or resolve disputes, as payments occur directly during pickup. Delivery services, route optimization, and navigation are also excluded, as transactions are limited to self-pickup. The system does not automatically detect or verify vegetable freshness because it relies entirely on seller-declared information, photographs, and listing details, without incorporating computer-vision freshness scanning, laboratory testing, or official inspections. The system does not enforce pricing or guarantee sales, as outcomes depend on user participation and market conditions. The weighted statistical tool is limited to ratings, reviews, and recorded reports within Agrivive and does not independently verify fraud or user claims. Analytics are limited to descriptive summaries of in-platform records, excluding external transactions and predictive forecasting. Beta-testing results apply only to selected participants and may not represent all markets in Davao City. System performance and marketplace activity may also be affected by factors beyond the researchers' control, such as limited or unavailable internet access, weak or unstable network signals, offline periods, hosting or third-party service interruptions, device capability, user engagement, data accuracy, weather conditions, and changes in market demand.

2. METHODS

This study used the Agile Software Development Methodology to guide the planning, analysis, design, development, testing, deployment, and evaluation of Agrivive. Agile follows iterative cycles where components are developed, tested, and improved based on the user's feedback [23], [24]. It was suitable for Agrivive because it involved multiple user groups whose input was needed to validate system processes. It also allowed the project to be divided into manageable sprints, enabling incremental testing and continuous refinement based on feedback.

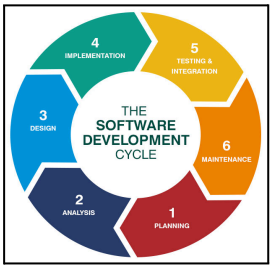

Figure 1. Agile Model

During the planning phase, the researchers defined the project objectives, target users, development schedule, technical requirements, and expected outputs. The requirements analysis phase involved surveys, interviews, observations, consultations, and a review of related systems to identify concerns involving limited surplus visibility, repeated manual promotion, changing buyer requirements, incomplete product and location information, unconfirmed pickup arrangements, trust concerns, and the absence of consistent record-keeping practices for surplus data. The validated requirements were then prioritized and grouped into development sprints according to the system's major components. During the system design phase, the researchers prepared the system architecture, database structure, user flows, interfaces, and process designs for the mobile and web applications. In the development phase, the approved designs were implemented through successive sprints covering account registration and verification, structured surplus listings, product search and filtering, marketing visibility, promotional content through n8n, seller-declared condition information, buyer-seller communication, notifications, current location pinpointing, QR-based reservations, reservation expiration, pickup confirmation, descriptive analytics, contextual advisories, and rule-based flagging and trust controls.

Each completed component proceeded to the testing phase, where its functionality and integration with other modules were examined. Testing included authentication, listing and quantity updates, QR-code generation and scanning, reservation status changes, automatic expiration, pickup confirmation, notifications, analytics records, and user-access permissions. After functional and integration testing, the completed prototype entered the deployment and beta-testing phase, during which representative sellers, buyers, and stakeholders evaluated the system in relation to the established requirements. In the evaluation and feedback phase, observations, test results, and user comments were reviewed to identify usability issues, functional errors, and areas requiring refinement. The findings were then incorporated into the next iteration, allowing affected components to return to the appropriate phase for revision, retesting, and further evaluation until the system reached its intended beta version [23], [24].

2.1 Planning

2.1.1 Data Gathering

The data-gathering activities involved 82 participants from Davao City. Of these, 79 participated in the survey, consisting of 38 produce sellers and 41 buyers. The sellers included suppliers, retail vendors, and supplier-vendors who distributed vegetables to regular or bulk buyers while also selling directly to walk-in customers. They were selected because of their direct experience in sourcing, handling, selling, and managing remaining vegetable quantities. The buyers consisted of household consumers and food-related businesses that purchased vegetables for consumption, resale, or food-service operations. They were selected because their changing product, quantity, price, condition, and pickup requirements directly affect vegetable sales trends. Three key informants representing the City Agriculturist's Office, City Economic Enterprises Office, and DFTC Vendor-Farmers Association were also included because of their knowledge of agricultural support, public-market administration, vegetable distribution, pricing practices, seller operations, and available marketplace

records. Data was gathered from Bankerohan Public Market, Agdao Public Market, and the Davao Food Terminal Complex. The researchers used surveys, face-to-face semi-structured interviews, and direct observations. Separate seller and buyer questionnaires were administered through printed forms and Google Forms. The seller questionnaire contained 21 questions and gathered information on the causes and occurrence of surplus, remaining quantities, current selling and promotional strategies, communication methods, pricing practices, product condition, location concerns, seller verification, and willingness to use a digital marketplace. The buyer questionnaire contained 20 questions and collected information on purchasing frequency and quantity, factors affecting buying decisions, preferred selling units, product-information needs, seller-verification preferences, pickup arrangements, location concerns, and willingness to purchase marketable surplus vegetables. Additional semi-structured questions were asked during survey administration to clarify responses and obtain examples of actual seller and buyer practices.

Face-to-face semi-structured interviews were conducted with representatives of CAgrO, CEEO, and the DFTC Vendor-Farmers Association on July 3 and July 11, 2026 using prepared interview guides. The interviews examined existing vegetable-market operations, surplus identification and handling, pricing and discount practices, seller-buyer coordination, verification procedures, market monitoring, and the availability of records on surplus quantities and buyer sales trends. Direct observations were also conducted during the market visits and survey administration for approximately 3 visits. Using field notes, the researchers observed vegetable displays, seller-buyer interactions, communication and promotional practices, handling of remaining vegetables, and accessibility of seller and pickup locations. The gathered data were used to define and validate Agrivive's functional and non-functional requirements. Seller findings informed the structured surplus listings, marketing and promotional features, seller-declared product information, communication tools, reservation and pickup processes, location support, and descriptive analytics. Buyer findings informed product searching and filtering, recent photograph and product-information requirements, seller-verification information, quantity selection, self-pickup arrangements, location details. The findings are limited to the selected participants and study sites and should not be interpreted as representing all vegetable sellers, buyers, organizations, or public markets in Davao City.

2.2 Analysis

2.2.1 User Requirements

Based on the data-gathering findings, Agrivive has three user roles: sellers, buyers, and stakeholders. Sellers include verified suppliers, supplier-vendors, and retail vendors who use the mobile application to manage and promote marketable surplus vegetables. Buyers include households, food-related businesses, resellers, and participating sellers purchasing from other sellers through the web application. Stakeholders include representatives of the City Agriculturist's Office, City Economic Enterprises Office, and other local offices. They have the most restricted access because they may only view aggregated and non-personally identifiable marketplace information.

2.2.1.1 Sellers shall be able to register, manage their account information, and verify their email address through OTP authentication.
2.2.1.2 Sellers shall be able to create, update, and archive surplus vegetable listings containing the product name, available quantity, selling unit, price, recent images, declared condition, storage condition, availability, and pickup information in order to present accurate details to potential buyers.
2.2.1.3 Sellers shall be able to view the visibility status of their active listings and access promotional captions and shareable links to promote remaining vegetables to potential buyers.
2.2.1.4 Sellers shall be able to send and receive in-app messages from buyers and receive notifications about new messages, reservation notices in order to respond to inquiries and manage self-pickup transactions.
2.2.1.5 Sellers shall be able to view available and reserved quantities in order to avoid accepting transactions that exceed the remaining stock.
2.2.1.6 Sellers shall be able to scan and validate a buyer's active QR code and confirm the product handover in order to complete a pickup transaction and record the quantity sold.
2.2.1.7 Sellers shall be able to view summaries of quantities posted, reserved, sold, cancelled, expired, and remaining, including sell-through rates and recurring surplus records, in order to review their marketplace activity and identify repeated overstock.
2.2.1.8 Sellers shall be able to respond to ratings, reviews, and feedback in order to participate in the marketplace feedback and trust process.
2.2.1.9 Buyers shall be able to register, browse, search, and filter available surplus vegetable listings in order to find products that match their preferred type, price range, and location.
2.2.1.10 Buyers shall be able to view product images, seller-declared condition information, available quantities, prices, pickup instructions, and seller-location details before arranging a purchase.
2.2.1.11 Buyers shall be able to send and receive in-app messages with sellers and receive notifications about new messages and replies in order to ask questions about products and coordinate purchases and self-pickup arrangements.
2.2.1.12 Buyers shall be able to follow sellers and receive notifications about promotions and price changes from the sellers they follow.
2.2.1.13 Buyers shall be able to select an available quantity and confirm a reservation in order to temporarily secure the requested vegetables for self-pickup.
2.2.1.14 Buyers shall be able to receive a QR code valid for 24 hours, view its remaining validity period and reservation status, cancel an eligible reservation, and present the QR code during pickup in order to complete the reservation process.
2.2.1.15 Buyers shall be able to view current location pinpoints, pickup instructions in order to locate the seller before travelling.
2.2.1.16 Buyers shall be able to rate completed transactions and the seller, provide reviews and feedback in order to participate in the marketplace feedback and trust process.
2.2.1.17 The public and stakeholders shall be able to view summarized marketplace information without personal information to monitor marketplace activity and planning.

2.2.2 Hardware and Software Requirements

The researchers identified the necessary specifications for the hardware and software used in the development and operation of the Agrivive system. The two tables below present the software tools used to develop the system and the hardware required for system development, deployment, and user access.

2.2.2.1 Hardware Requirements

Table 2. Hardware Requirements

Hardware	Minimum Specification	Purpose
Development Computer	Intel Core i5 / AMD Ryzen 5 or equivalent	Used by the researchers for system development, debugging, testing, deployment, and maintenance
RAM	Minimum 8 GB	Supports software development and multitasking
Storage	256 GB SSD	Stores source code, database files, and development tools
Display	1366 × 768 resolution	Used for developing and testing the user interface
Mobile Device	Android 10 or latest, Quad-Core Processor, Minimum 4 GB RAM	Used by sellers surplus-listing management, buyer communication, QR-code scanning, marketing-content review, advisories, and supplier analytics.
Desktop/Laptop	Intel Core i3 / AMD Ryzen 3 or equivalent, Minimum 4 GB RAM, 1366 × 768 display, Modern Web Browser	Used by buyers, administrators, and stakeholders to access the Agrivive web application.
Tablet	Android 10 or later, Minimum 3 GB RAM, Modern Web Browser	Used by buyers and stakeholders to access the Agrivive web application.
Hostinger Virtual Private Server	At least 2 vCPU cores, 4 GB RAM, and 50 GB SSD storage	Hosts the web application, backend services, PostgreSQL database, Docker containers, and n8n workflows.
Internet Connection	Minimum 10 Mbps	Needed to use the app, website, messaging, and to sync your transactions in real time

The hardware requirements outlined in Table 2 provide the minimum resources needed to develop, test, deploy, and operate Agrivive. Sellers primarily use mobile devices, while buyers, administrators, and stakeholders may access the web application through computers or tablets. A stable internet connection and Hostinger VPS are required to support system hosting, data synchronization, mapping, notifications, and n8n workflows.

2.2.2.2 Software Requirements

Table 3. Software Requirements

Software	Version	Usage
NextJS	16.3.4	Develops the buyer, administrator, and stakeholder web application.
React QR Code	2.2.0	Generates the buyer's 24-hour reservation QR code.
React Native	0.86	Develops the supplier and vendor mobile application.
React Native QR Scanner Library	5.1.0	Scans buyer QR codes for reservation and pickup confirmation.
Groq SDK/API	1.3.0	Converts structured analytics records into readable summaries without forecasting future demand
ElysiaJS	1.4.29	Handles backend APIs, listings, reservations, and transactions. Manages the process of generating text summaries from descriptive analytics.
Tailwind CSS	4.3.2	Styles the responsive web interface.
Docker	29.6.1	Containerizes the application, database, and automation services.
Web browser	Chrome, Edge, Firefox, or Safari	Used by Buyers, Administrator, and other stakeholders to access and interact with the Agrivive web application.
Better Auth	1.6.23	Manages authentication, sessions, and protected access.
PostgreSQL	18.4	Stores user, listing, reservation, transaction, and analytics data.
Amazon S3	Latest	Stores uploaded product images
Hostinger	v3	Hosts the application, backend, database, and deployed services.
n8n	2.30.5 Stable	Automates marketing-content preparation and configured marketing alerts only.
OpenStreetMap	API 0.6 with Standard Tile Layer	Supplies the base map and geographic information.
React Leaflet	5.0.0	Displays interactive seller and pickup-location maps.
Nager.Date API	API v4	Supplies Philippine public-holiday information for calendar advisories.

The software tools outlined in Table 3 support Agrivive's mobile and web interfaces, backend services, authentication, data storage, mapping, workflow automation, external data references, and descriptive analytics. Together, these technologies provide the necessary environment for developing and maintaining the cross-platform marketplace.

2.2.3 Functional and Non-Functional Requirements

2.2.3.1 Functional Requirements

Table 4. Functional Requirements

User Role	Module	FR No.	The System Shall...
All registered users	Authentication & Access	FR-01	Allow account creation, authenticate users through OTP verification, enable secure logout, and enforce role-based access to protected functions.
Sellers	Registration	FR-02	Create and manage their account information and verify their email address through OTP.
	Listing & Inventory Management	FR-03	Create, update, and archive listings and manage product details and quantities. Track available, reserved, sold, and remaining quantities.

	Communication & Notifications	FR-04	Receive buyer messages, and reservation notices.
	Marketing Visibility	FR-05	Display listing visibility levels and allow sellers to view promotional captions and shareable links.
	Reservation & QR Processing	FR-06	View reservations, scan and validate QR codes, and confirm product handover.
	Analytics & Advisories	FR-07	Display seller analytics, sell-through rates, and surplus records.
	Trust & Accountability	FR-08	Allow sellers to view and respond to ratings, reviews, and feedback from buyers.
Buyers	Marketplace Discovery & Evaluation	FR-09	Display listings and allow buyers to search and filter by product type, price range, and location.
	Communication & Notifications	FR-10	Allow buyers to message sellers on product listings and receive seller replies. Allow buyers to subscribe to sellers and receive notifications about promotions and price changes.
	Reservation	FR-11	Allow buyers to select available quantities and confirm or cancel eligible reservations.
	QR Reservation	FR-12	Generate a QR code valid for 24 hours and display its status and remaining validity period.
	Location & Pickup	FR-13	Display current location, distance, and pickup instructions.
	Ratings, Reviews & Reporting	FR-14	Allow buyers to rate the product and seller and provide reviews and feedback.
Stakeholders	Aggregated Data Access	FR-15	Allow stakeholders to view summarized marketplace information without personal information.

The functional requirements outlined in Tables 4. covers all user roles and documents both user-initiated and system-triggered behaviors. Each requirement can be evaluated through a corresponding functional test case during system testing and beta evaluation.

2.2.3.2 Non-Functional Requirements

Table 5. Non-Functional Requirements

Category	NFR No.	The System Shall...
Performance	NFR-01	Load primary pages within an acceptable response time under normal operating conditions.
	NFR-02	Display listing search and filter results within an acceptable response time after a valid request under normal operating conditions.
	NFR-03	Validate and display the status of a scanned reservation QR code within an acceptable response time under normal network conditions.
	NFR-04	Generate a requested descriptive analytics summary within acceptable response time for records available during beta testing.
Security	NFR-05	Require successful authentication and OTP verification before granting access to protected account functions.
	NFR-06	Enforce role-based authorization on protected pages, application functions, and backend API requests.
	NFR-07	Store user passwords only through the secure hashing mechanism provided by the implemented authentication service.
	NFR-08	Transmit authenticated user data through HTTPS in the deployed environment.
Usability	NFR-09	Provide responsive mobile interfaces at a minimum supported width of 360 pixels and responsive web interfaces at a minimum resolution of 1366 × 768.
	NFR-10	Display clear error messages that tell users what went wrong and what they need to do.
	NFR-11	Provide simple and consistent navigation for listing search, filtering, reservation, QR pickup, and product discovery.
Reliability	NFR-12	Keep confirmed user, listing, reservation, transaction, feedback, and analytics records after a page refresh, logout, or device reconnect.
	NFR-13	Prevent available, reserved, sold, and remaining quantities from becoming negative after listing or reservation actions.
	NFR-14	Keep quantity totals correct after reservation, cancellation, expiration, and pickup confirmation.
Availability	NFR-15	Remain accessible during scheduled beta-testing periods except during announced maintenance, hosting interruptions, or third-party service outages.
Compatibility	NFR-16	Operate on supported seller devices running Android 10 or later.
	NFR-17	Provide the core web functions on the current stable releases of Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.
Maintainability	NFR-18	Use modular mobile, web, backend, database, analytics, and automation components so that one module can be corrected or updated without requiring changes to unrelated modules.
	NFR-19	Maintain documented configuration values for authentication, database access, external APIs, hosting, and automation workflows to support deployment and testing.
Data Integrity	NFR-20	Validate required fields, data types, quantity values, prices, coordinates, and uploaded-file formats before storing or updating a record.
	NFR-21	Prevent the same reservation or QR code from being confirmed more than once.
	NFR-22	Perform reservation, cancellation, expiration, and pickup quantity changes as complete database transactions so that partial updates are not stored.

The non-functional requirements in Table 5 establish verifiable quality standards appropriate to Agrivive's cross-platform and transaction-based environment. The requirements emphasize performance, secure OTP authentication, privacy, usability, reliable quantity management, compatibility, modularity, analytical processing, and network resilience. They also ensure that the weighted statistical tool processes ratings, reviews, feedback, and transaction-related reports consistently. These requirements support reliable system operation even when network connectivity is weak or temporarily unavailable.

2.2.4 Feasibility Study

The feasibility study assesses whether Agrivive can be developed, deployed, operated, and maintained within the available user environment, technologies, financial resources, and Academic Year 2026–2027 schedule. The assessment is based on the surveys, interviews, observations, user and system requirements, selected development tools, and planned project activities.

2.2.4.1 Operational Feasibility

Agrivive is operationally feasible because its functions reflect the existing practices and concerns of sellers, buyers, market representatives, and government stakeholders in the selected Davao City sites. Local sellers may continue using physical stalls, regular buyers, personal contacts, and social-media promotion while using the system as an additional channel for posting surplus vegetables and coordinating self-pickup transactions. Survey findings showed that 94.7% of the sellers were willing to use an application that could help them reach alternative buyers, while all 41 buyers were willing to use self-pickup to avoid delivery fees. Buyers also identified the need for recent product photographs, seller legitimacy information, updated availability, and identifiable pickup locations, which correspond with the system requirements. However, adoption may still be affected by limited digital literacy, reliance on familiar practices, incomplete user-submitted information, and inconsistent participation; therefore, orientation and system demonstrations be provided during beta testing. Based on the identified workflows and expressed user willingness, Agrivive is operationally feasible within the selected study setting, although acceptance cannot be assumed for all vegetable sellers and buyers in Davao City.

2.2.4.2 Technical Feasibility

Agrivive is technically feasible because the hardware, software, hosting services, and development tools required for its beta version are available and correspond with the requirements identified in Section 2.2.2. Next.JS and React Native support the web and mobile interfaces, while ElysiaJS, PostgreSQL, Better Auth, Tailwind CSS, and Docker support backend processing, data storage, authentication, interface development, and deployment. React Leaflet and OpenStreetMap provide seller-location functions, while Nager.Date supplies external Philippine public-holiday references. Hostinger VPS provides the deployment environment, n8n is limited to configured marketing-content and notification workflows, and Python supports ElysiaJS with Groq converts validated descriptive analytics into readable summaries without calculating new statistics or forecasting demand. The selected technologies can support accounts, surplus listings, reservations, QR-code confirmation, marketing visibility, location information, transaction records, and descriptive analytics within the intended beta-testing scale. Although internet instability, device limitations, hosting interruptions, external-service outages, and integration issues may affect performance, these risks do not prevent development and testing; therefore, Agrivive is technically feasible provided that fallback handling, testing, and configuration management are applied.

2.2.4.2 Economic Feasibility

Agrivive is economically feasible for a student capstone prototype because most of its primary frameworks and libraries are open-source and do not require software-licensing fees. Its estimated costs consist mainly of development labor value, internet use, electricity, documentation, initial hosting, annual hosting renewal, and maintenance. Open-source tools still involve indirect costs because they require development time, hardware use, connectivity, testing, deployment, and technical maintenance. The expected benefits are directly connected to the identified problems, including broader visibility of marketable surplus, reduced dependence on repeated messages, clearer pickup coordination, organized transaction records, and descriptive information about recurring surplus. However, alternative selling opportunities represent potential benefits rather than guaranteed income because actual sales remain dependent on sales trend, pricing, condition, location, and participation. Consequently, Agrivive is economically feasible for development and beta deployment, subject to verification of the final Hostinger, maintenance, domain, and usage-based service costs before full operation.

Table 6. Cost and Benefit Analysis

Item	Type	Estimated Cost
Development Costs		
Programmer / Developer Labor (6-month development)	Direct	₱180,000.00
Internet/Data Usage (6 months)		₱1,500.00
Electricity (6 months)		₱2,400.00
Documentation and Printing		₱1,000.00
Open-source Licensing		₱0.00
Hostinger VPS hosting (1 month)		₱409.00
External services during beta testing under available free tiers		₱0.00
Total Estimated Development Cost		**₱185,309.00**
Estimated Annual Operational Costs		
Hostinger VPS hosting at ₱409.00 per month	Direct	₱4,908.00
System Maintenance and Minor Updates		₱15,000.00
Estimated Annual Operational Cost		**₱19,908.00**
Expected Benefits	**Type**	**Expected Impact**
Expected Benefits		
Increased visibility of surplus listings	Intangible	Wider access to potential buyers
Seller-location pinpointing		Easier identification of pickup locations
Descriptive surplus analytics		Improved monitoring of recorded surplus patterns
Reduced repeated communication	Tangible	Lower time and communication effort
Organized transaction and reservation records		Reduced dependence on manually compiled marketplace records
Alternative selling opportunities	Potential tangible	Opportunity to recover part of the value of marketable surplus

Table 6 presents the expected development and operational expenses of Agrivive together with its potential benefits. The final cost must be recalculated using verified Hostinger, domain, and external API quotations before system deployment.

2.2.4.4 Schedule Feasibility

Agrivive is schedule feasible because its activities are divided into phases and iterative sprints within Academic Year 2026–2027. Planning begins in June 2026, requirements analysis is conducted from June to July, system design occurs from July to August, and development is allocated the longer period from August to October. Functional and integration testing, beta evaluation, and revisions are scheduled from October to November, followed by deployment activities from November to December and maintenance beginning in December. The overlapping phases reflect Agile development, in which approved components may enter design, development, and testing while feedback on other components is still being evaluated. If delays occur, the team may reprioritize incomplete sprint tasks and retain only the functions included in the approved capstone scope rather than introducing additional features. Given the planned phase durations, iterative workflow, and inclusion of post-deployment maintenance, Agrivive is schedule feasible within the academic period, provided that development progress and dependencies are monitored regularly.

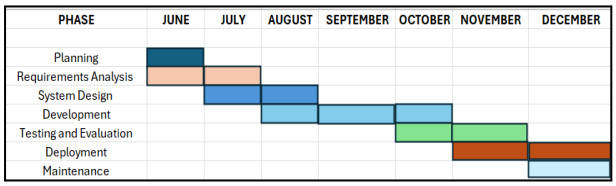

Figure 2. Gantt Chart

2.3 Design

2.3.1 Conceptual Framework

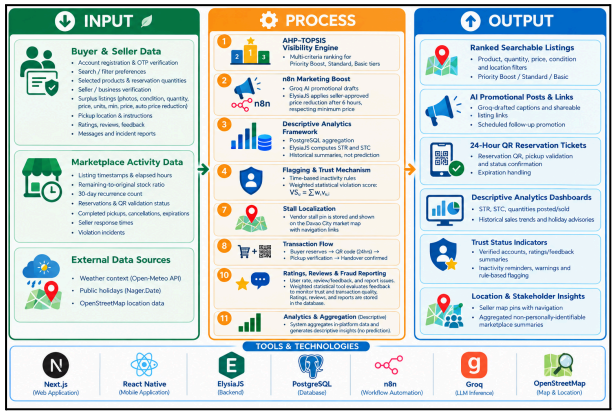

Figure 3. Conceptual Framework

Figure 3 presents the conceptual framework of Agrivive using the Input–Process–Output (IPO) model. The framework illustrates how operational data from buyers, vegetable suppliers, retail vendors, marketplace activities, and external services are collected, processed, and transformed into ranked marketplace listings, automated promotional outputs, QR-verified transaction records, descriptive sales-trend insights, and trust-monitoring actions. Agrivive integrates four primary mechanisms within a localized Davao City marketplace: the AHP–TOPSIS Multi-Criteria Decision Visibility Engine, the n8n Marketing Boost Mechanism, the Descriptive Supplier Sales Trend Analytics Framework, and the Flagging and Trust Mechanism, alongside multi-criteria product search, 24-hour QR-based reservation and pickup confirmation, and seller geolocation. Together, these functions address limited surplus exposure, difficult product discovery, repeated manual promotion, uncertain pickup coordination, trust concerns, and the lack of organized historical records for surplus vegetable transactions. The system relies on rule-based and descriptive processing and does not perform predictive demand forecasting. Transaction ratings, reviews, user feedback, and violation reports are evaluated through time-based rules and a weighted statistical tool to support marketplace accountability. The system's overall effectiveness is assessed through functional testing, beta testing, and user evaluation.

2.3.1.1 Input

As shown in Figure 3, the Input component consists of user-submitted data entered through Agrivive's web and mobile applications, system-logged marketplace activity, and reference data retrieved from external APIs. Buyer inputs include account registration and OTP verification details, multi-criteria search and filter preferences, selected products and reservation quantities, seller subscriptions, in-app messages, transaction ratings, reviews, feedback, and incident or fraud reports. Supplier and vendor inputs include account and business verification credentials, surplus vegetable listings, recent product photographs, seller-declared condition details, original and available quantities, selling units, initial prices, minimum price floors and automatic price-reduction percentages, pickup instructions, and geolocation pinpoints. Marketplace activity data include listing publication timestamps, elapsed posting hours, remaining-to-original stock ratios, 30-day surplus recurrence counts, reservation timestamps, 24-hour QR-code validation statuses, completed pickups, cancellations, expirations, seller response times, and recorded violation incidents. External inputs include Philippine public-holiday data from the Nager.Date API for contextual advisories and base-map tiles from OpenStreetMap for location pinpointing.

2.3.1.2 Process

The Process component transforms collected inputs into core services and analytical outputs using Next.js, React Native, ElysiaJS, Better Auth, and PostgreSQL. Listing prioritization is executed via the AHP–TOPSIS Multi-Criteria Decision Visibility Engine, evaluating Listing Recency (C_1), Stock Availability (C_2), Seller Reliability (C_3), and Recurrence (C_4). Qualified Priority Boost listings trigger the n8n Marketing Boost Mechanism, leveraging Groq-hosted models for AI promotional drafting and ElysiaJS for automated 6-hour price-reduction workflows. The reservation workflow handles stock locking, 24-hour QR code generation, automatic stock restoration, and React Leaflet map rendering. Historical performance is evaluated by the Descriptive Supplier Sales Trend Analytics Framework, computing Sell-Through Rates (STR) and Sales Trend Changes (STC) backed by Groq summaries and Nager.Date advisories. Finally, the Flagging and Trust Mechanism combines time-based seller inactivity rules with a weighted statistical violation score model

($VS_u = \sum w_i v_{u,i}$) to determine account actions.

2.3.1.3 Output

Figure 3 shows the outputs generated by Agrivive across its user roles. These include verified seller and buyer accounts, ranked surplus vegetable listings with assigned visibility tiers (Priority Boost, Standard Exposure, and Basic/Hold), and

multi-criteria filtered search results. Promotional outputs include AI-drafted promotional captions, shareable listing links, scheduled price-adjusted follow-up posts, and automated buyer notifications. Transaction and coordination outputs include interactive seller pickup maps, 24-hour reservation QR codes, real-time inventory updates across available, reserved, and sold quantities, automatic stock restoration for expired or cancelled reservations, and completed pickup records. Analytical outputs include seller dashboards displaying sell-through rates, period-to-period sales trend changes, frequently posted surplus items, readable descriptive summaries, calendar-based holiday advisories, and aggregated non-personally identifiable marketplace summaries for stakeholders. Trust and accountability outputs include verified transaction ratings, weighted violation evaluation summaries, automated inactivity reminders and warnings, and rule-based account status updates. Together, these outputs support product discovery, surplus visibility, pickup coordination, automated promotion, and marketplace monitoring without guaranteeing sales, certifying food safety or internal freshness, or predicting future buyer demand.

2.3.2. Integrated Surplus Visibility, Marketing, Analytics, and Trust Framework

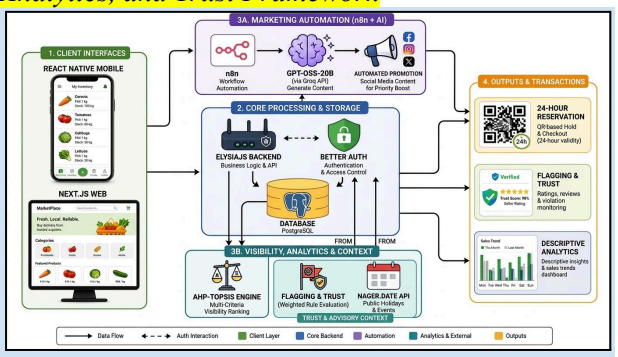

Figure 4. Pipeline

The primary Capstone Element of Agrivive is the Integrated Surplus Visibility, Marketing, Analytics, and Trust Framework, a rule-based and data-driven framework composed of four connected components: the AHP–TOPSIS Multi-Criteria Decision Visibility Engine, the n8n Marketing Boost Mechanism, the Descriptive Supplier Sales Trend Analytics Framework, and the Flagging and Trust Mechanism. Together, these components evaluate and rank active surplus listings to assign visibility tiers, automate promotional content drafting and scheduled price-reduction workflows, summarize historical marketplace performance through sell-through rates and period-to-period sales trends, and support transaction accountability through time-based rules and weighted statistical evaluation.

The framework considers factors related to marketable surplus visibility, listing recency, remaining stock availability, seller reliability, historical surplus recurrence, promotional outreach, completed QR-validated pickup records, and transaction-related trust concerns such as misleading product information, cancellations, no-shows, and non-responsive users. Agrivive applies configured rules and analytical processes to listing, reservation, QR-validated pickup, sales-trend, and user-activity records. Specifically, the AHP–TOPSIS Multi-Criteria Decision Visibility Engine determines listing priority to trigger automated marketing workflows in n8n, the Descriptive Supplier Sales Trend Analytics Framework aggregates completed pickup transactions into historical performance summaries, and the Flagging and Trust Mechanism applies time-based thresholds and a weighted statistical tool to evaluate recorded violations and determine corresponding account actions. However, the framework does not independently determine actual vegetable freshness, certify food safety, predict future buyer demand, guarantee completed sales, or treat unverified reports as confirmed violations.

2.3.2.1 AHP–TOPSIS Multi-Criteria Decision Visibility Engine

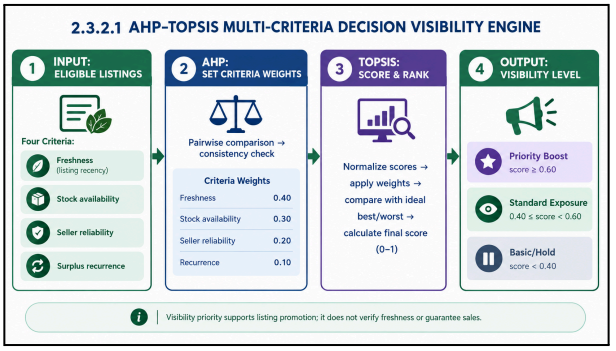

Figure 5. AHP–TOPSIS Multi-Criteria Decision Visibility Engine

The AHP–TOPSIS Multi-Criteria Decision Visibility Engine is a method used by Agrivive to rank active surplus vegetable listings and assign their visibility levels. The Analytic Hierarchy Process (AHP) determines how important each criterion is by comparing them in pairs and checking whether the comparisons are consistent [1]. The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) ranks listings by comparing their results with the best and worst possible results [2].

The engine evaluates four criteria: Listing Recency(C1), Stock Availability (C2), Seller Reliability (C3), and Historical Surplus Recurrence (C4). AHP determines the weight of each criterion, while TOPSIS calculates a score for each listing using these weights and the listing's data [1], [2]. Similar methods have been used to evaluate farming practices [3], study TOPSIS applications in agriculture [4], select data analytics platforms for precision farming [5], and identify suitable agricultural land [6].

The final score determines the listing's visibility level: Priority Boost for scores of 0.60 and above, Standard Exposure for scores from 0.40 to below 0.60, and Basic/Hold for scores below 0.40. Priority Boost listings may receive better placement in Agrivive and qualify for promotional activities through n8n. The weights and score limits are settings created for the Agrivive prototype, not universal agricultural standards. The score only determines promotional priority; it does not confirm vegetable freshness or safety, or guarantee that a listing will be sold.

The AHP–TOPSIS Multi-Criteria Decision Visibility Engine evaluates active surplus vegetable listings based on four criteria: Listing Recency, Stock Availability, Seller Reliability, and Historical Surplus Recurrence. The process begins by checking listing eligibility and calculating the score for each criterion. AHP then uses pairwise comparisons to determine

the weight of each criterion and checks the consistency of the comparisons [1]. The resulting weights are applied in TOPSIS, which normalizes the listing data, identifies the ideal best and worst results, and calculates the relative closeness score for each listing [2]. The final scores are used to rank listings and assign their visibility levels according to Agrivive's configured thresholds. Listings classified as Priority Boost may qualify for automated promotional activities through n8n. This process helps Agrivive apply the same evaluation method to eligible listings while supporting consistent decisions about promotional visibility.

Stage 1: Criterion Scores
Each eligible listing is evaluated using four criteria. The following formulas convert the recorded listing data into scores ranging from 0 to 1. These scores are used to compare listings during the TOPSIS process.

1. Listing Recency (C1)
The system uses the time since publication as an indicator of listing recency. It does not directly measure the actual freshness of the vegetables.

$$[C_{i1} = 1 - \min\left(\frac{hi}{12},\ 1\right)]$$

Where:

- (C_{i1}) = listing recency score of listing (i);
- (h_i) = number of hours elapsed since listing publication;
- (12) = configured reference period in hours.

Where (h_i) is the number of hours since publication and 12 is the configured reference period. A newly published listing receives a score of 1.00, while a listing that has been published for 12 hours or longer receives a score of 0.00.

2. Stock Availability (C2)
This criterion measures how much of the original listed quantity remains available.

$$[C_{i2} = \min\left(\frac{qr,i}{qo,i},\ 1\right)]$$

Where:

- (C_{i2}) = stock availability score of listing (i);
- $(q_{r,i})$= current remaining quantity;
- $(q_{o,i})$ = originally posted quantity.

Where $(q_{r,i})$ is the current remaining quantity and $(q_{o,i})$ is the originally posted quantity. A higher score means that a larger portion of the original stock remains available.

3. Seller Reliability (C3)
This criterion uses the seller's recorded transaction ratings to calculate a normalized score.

$$C_{i3} = \begin{cases} \frac{\bar{ri}}{5}, & \text{if valid rating exist} \\ 0.50, & \text{if no rating history exists} \end{cases}$$

Where:

- (C_{i3}) = seller reliability score associated with listing (i);
- $(\bar{r}_i)$ = average recorded transaction rating for the seller;
- (5) = maximum rating on the five-star scale.

Where $(\bar{r}_i)$ is the seller's average recorded transaction rating and 5 is the maximum five-star rating. If no rating history exists, the system assigns a baseline score of 0.50. This score represents recorded ratings rather than independently verified seller behavior.

4. Historical Surplus Recurrence (C4)
This criterion measures how often the same seller has previously listed the same type of vegetable as surplus within the preceding 30 days.

$$[C_{i4} = \min\left(\frac{Fi}{3},\ 1\right)]$$

Where:

- (C_{i4}) = historical surplus recurrence score of listing (i);
- (F_i) = number of previous qualifying surplus listings within the observation period;
- (3) = configured maximum recurrence reference.

Where (F_i) is the number of previous qualifying surplus listings and 3 is the configured maximum reference. The score is 0.00 for no previous listings, approximately 0.33 for one, 0.67 for two, and 1.00 for three or more previous listings.

Stage 2: AHP Criterion Weights
AHP determines how much importance is assigned to each of the four criteria. It uses a pairwise comparison matrix to compare the criteria and calculate their relative weights [1].
The comparison matrix used in the Agrivive prototype is:

$$A = \begin{bmatrix} 1.000 & 1.333 & 2.000 & 4.000 \\ 0.750 & 1.000 & 1.500 & 3.000 \\ 0.500 & 0.667 & 1.000 & 2.000 \\ 0.250 & 0.333 & 0.500 & 1.000 \end{bmatrix}$$

The rows and columns follow the order C1, C2, C3, and C4. Each value represents the relative importance of one criterion compared with another.

The geometric mean of each row is calculated as follows [1]:

$$[g_j = \left(\prod_{k=1}^{4} ajk\right)^{1/4}]$$

The normalized weight of each criterion is then calculated by dividing its geometric mean by the sum of all four geometric means:

$$[w_j = \frac{g_j}{\sum_{r=1}^{4} g_r}]$$

The resulting weights are approximately:

$$W = \begin{bmatrix} 0.400 \\ 0.300 \\ 0.200 \\ 0.100 \end{bmatrix}$$

Criterion	Weight
Listing Recency (C1)	0.400
Stock Availability (C2)	0.300
Seller Reliability (C3)	0.200
Recurrence (C4)	0.100

The weights must sum to 1:

$$\sum_{j=1}^{4} wj = 1$$

AHP Consistency Check

AHP checks whether the pairwise comparisons are sufficiently consistent using the maximum eigenvalue, Consistency Index (CI), and Consistency Ratio (CR) [1].

The maximum eigenvalue is estimated as:

$$\lambda_{\max} = \frac{1}{4}\sum_{j=1}^{4}\frac{(Aw)j}{Wj}$$

The Consistency Index is calculated as:

$$CI = \frac{\lambda max - n}{n-1}$$

The Consistency Ratio is calculated as:

$$CR = \frac{CI}{RIn}$$

Where $(n = 4)$ is the number of criteria and $(RI_4 = 0.90)$ is the Random Index for four criteria. A CR below 0.10 is the commonly used threshold for acceptable consistency [1]. The comparison matrix used in the prototype produces a CR close to zero, indicating that its pairwise comparisons are consistent.

Stage 3: TOPSIS Ranking

After the weights are determined, TOPSIS compares the eligible listings based on their criterion scores and distances from the ideal best and worst results [2].

Let $(X = [x_{ij}])$ represent the decision matrix, where (n) is the number of eligible listings, (m=4) is the number of criteria, and (x_{ij}) is the score of listing (i) under criterion (j).

1. Vector Normalization

The values in each criterion column are normalized using the following formula [2]:

$$rij = \frac{xij}{\sqrt{\sum_{i=1}^{n} x^2 ij}}$$

Where:

- (r_{ij}) = normalized value of listing (i) under criterion (j);
- (x_{ij}) = operational criterion value of listing (i);
- (n) = number of eligible listings being compared.

This step converts the criterion values into comparable proportions across the eligible listings.

2. Weighted Normalization

Each normalized value is multiplied by its corresponding AHP weight [1], [2]:

$$V_{ij} = W_j r_{ij}$$

Where:

- (v_{ij}) = weighted normalized value of listing (i) under criterion (j);
- (w_j) = AHP weight for criterion (j);
- (r_{ij}) = vector-normalized criterion value.

Where (v_{ij}) is the weighted normalized value, (w_j) is the criterion weight, and (r_{ij}) is the normalized value.

3. Ideal Best and Ideal Worst Solutions

TOPSIS identifies the best and worst reference values for each criterion by finding the highest and lowest weighted normalized values [2].

$$A^+ = \left(\max_i v_{i1}, \ldots, \max_i v_{i4}\right)$$

The ideal worst solution is:

$$A^- = \left(\min_i v_{i1}, \ldots, \min_i v_{i4}\right)$$

Where:

- (A^+) = ideal best solution;
- (A^-)= ideal worst solution;
- (v_{ij}) = weighted normalized value of listing (i) under criterion (j).

These solutions serve as reference points for comparing the eligible listings [2].

4. Euclidean Distances

The distance of each listing from the ideal best solution is:
The distance from the ideal best solution is:

$$D_i^+ = \sqrt{\sum_{j=1}^{4}(V_{ij} - V_j^+)^2}$$

The distance from the ideal worst solution is:

$$D_i^- = \sqrt{\sum_{j=1}^{4}(V_{ij} - V_j^-)^2}$$

Where:

- (D_i^+) = distance from the ideal best solution;
- (D_i^-) = distance from the ideal worst solution;
- (v_j^+) = ideal best value for criterion (j);
- (v_j^-) = ideal worst value for criterion (j).

These formulas measure how far each listing is from the best and worst reference points [2].

5. Relative Closeness Coefficient

The final TOPSIS score is calculated using the two distances [2]:

$$C_i^* = \frac{D_i^-}{D_i^+ + D_i^-}$$

Where:

- (C_i^*) = relative closeness coefficient of listing (i);

- (D_i^+) = distance from the ideal best solution;
- (D_i^-)= distance from the ideal worst solution.

The coefficient lies within the interval:

$0 \leq C_i^* \leq 1$

Where (C_i^*) is the relative closeness coefficient of listing (i).

The score ranges from 0 to 1. A higher score means the listing is closer to the ideal best solution and farther from the ideal worst solution within the evaluated group.

Stage 4: Visibility Classification

Agrivive uses the final TOPSIS score to assign each eligible listing to a visibility level. The thresholds below are configurations defined for the prototype.

$$\text{Tier}(C_i^*) = \begin{cases} \text{Priority Boost}, & C_i^* \geq 0.60 \\ \text{Standard Exposure}, & 0.40 \leq C_i^* < 0.60 \\ \text{Basic/Hold}, & C_i^* < 0.40 \end{cases}$$

Listings are ranked from the highest score to the lowest. If listings have equal scores, publication timestamps and then unique listing identifiers are used to determine their order. Priority Boost listings may qualify for promotional activities through n8n, while the visibility level itself does not confirm vegetable freshness, food safety, or guaranteed sales.

Sample Computation

Consider three eligible surplus vegetable listings with the following illustrative criterion scores. All values have already been converted to the range from 0 to 1.

Listing	Freshness	Stock Availability	Seller Reliability	Recurrence
A	0.80	0.75	0.90	0.67
B	0.80	0.50	0.80	0.33
C	0.40	0.80	0.60	1.00

Using the AHP weights, TOPSIS calculates the weighted normalized values, ideal solutions, and distances for each listing [1], [2].

For Listing A, the calculated distances from the ideal best and ideal worst solutions are approximately:

$$[D_A^+ = 0.0292, \quad D_A^- = 0.1693]$$

The relative closeness coefficient is:

$$[C_A^* = \frac{0.1693}{0.0292 + 0.1693}]$$

$$[C_A^* \approx 0.853]$$

Since the score is greater than or equal to 0.60, Listing A receives **Priority Boost**.

The resulting scores and visibility levels for all three listings are:

Listing	(D_i^+)	(D_i^-)	(C_i^{+*})	Visibility Level
A	0.0292	0.1693	0.853	Priority Boost
B	0.1191	0.0800	0.402	Standard Exposure
C	0.1551	0.0920	0.372	Basic/Hold

The sample demonstrates how TOPSIS converts multiple criterion scores into a final ranking score. Agrivive then uses that score to determine the visibility level of each eligible listing. These values are illustrative examples, not actual production results.

2.3.2.2 n8n Marketing Boost Mechanism

Figure 6. n8n Marketing Boost Mechanism

The Marketing Boost Mechanism in Agrivive uses n8n, a workflow automation tool, to coordinate the preparation and promotion of active surplus vegetable listings through supported online channels. n8n connects applications, databases, and services through a series of connected steps called nodes, allowing repetitive tasks to be automated through a visual interface with minimal manual coding. Marketing automation can support campaign performance and marketing efficiency [30], while social media provides additional channels for promoting agricultural products [31] and sharing information related to food-waste reduction [32]. In Agrivive, the mechanism helps suppliers and vendors increase the visibility of their marketable surplus vegetables by preparing promotional content, generating shareable listing links, sending notifications, and coordinating scheduled promotional activities. These activities provide additional opportunities for households, carinderias, restaurants, resellers, and other potential buyers to discover available surplus vegetables beyond the Agrivive platform. By making eligible listings easier to share and discover, the mechanism supports the redistribution of sellable surplus vegetables to potential buyers. However, increased visibility does not guarantee that buyers will view, reserve, or purchase the listed products.

The process begins when an active surplus vegetable listing is evaluated by Agrivive's background worker service in ElysiaJS, the backend framework. The backend verifies that the listing is from an active, verified seller, has available stock, and satisfies the required visibility threshold. Listing classification and priority scoring are calculated using Agrivive's Multi-Criteria Decision Visibility Engine (incorporating posting age, remaining quantity ratio, and recurrence). Once a listing is confirmed eligible for Priority Boost, ElysiaJS triggers the n8n workflow through an authenticated webhook. n8n then fetches the sanitized listing context from ElysiaJS, sends the information to openai/gpt-oss-20b (via Groq) to generate a neutral

promotional headline and caption, and submits the draft back to ElysiaJS for persistence. The seller-provided listing information remains unchanged unless an approved price update is processed through the backend. If direct automated social posting is unavailable due to integration or permission limitations, the prepared promotional copy and listing link remain accessible in the seller interface for manual sharing.

To prevent long-running, idle workflow executions in n8n, scheduled follow-up promotions are managed directly by ElysiaJS and PostgreSQL. After the initial promotional draft is stored, the backend schedules a follow-up promotion job set for six hours later. When the scheduled time arrives, the backend worker re-evaluates the listing to verify that it remains active, has available stock, and has not been deactivated. If the product is sold out or deactivated, the workflow stops. If the listing remains eligible and the seller has configured an automatic price reduction, ElysiaJS applies the scheduled percentage reduction toward the seller's minimum price:

$$P_{new} = \max(P_{current} \times (1 - r), P_{minimum})$$

Where P_{new} is the updated price, $P_{current}$ is the current price, r is the seller-approved reduction percentage, and $P_{minimum}$ is the minimum price floor set by the seller. ElysiaJS performs the calculation and updates the official listing record in PostgreSQL, while n8n coordinates the subsequent drafting workflow without directly calculating or modifying product prices.

Once ElysiaJS verifies the follow-up eligibility and any updated price, it dispatches a follow-up webhook to n8n. n8n sends the updated listing information back to openai/gpt-oss-20b to generate a follow-up promotional title and caption reflecting the current availability and pricing. The generated copy directs interested buyers to the Agrivive product listing for reservations. The language model is responsible strictly for drafting text; it does not determine listing eligibility, calculate official selling prices, or assign visibility tiers.

Finally, the Marketing Boost Mechanism updates the promotion job records in PostgreSQL to maintain a history of promotional actions performed for each surplus listing cycle. These records track the product identifier, stage (initial or followup), generated copy, completion timestamps, and delivery status. Each component maintains a distinct responsibility: n8n coordinates external workflow automation and AI drafting; ElysiaJS validates listing eligibility and handles approved price calculations; PostgreSQL stores authoritative listing information and job histories; and openai/gpt-oss-20b generates promotional text. Supported external platforms provide channels for sharing listings and directing interested buyers back to Agrivive.

2.3.2.3 Descriptive Supplier Sales Trend Analytics Framework

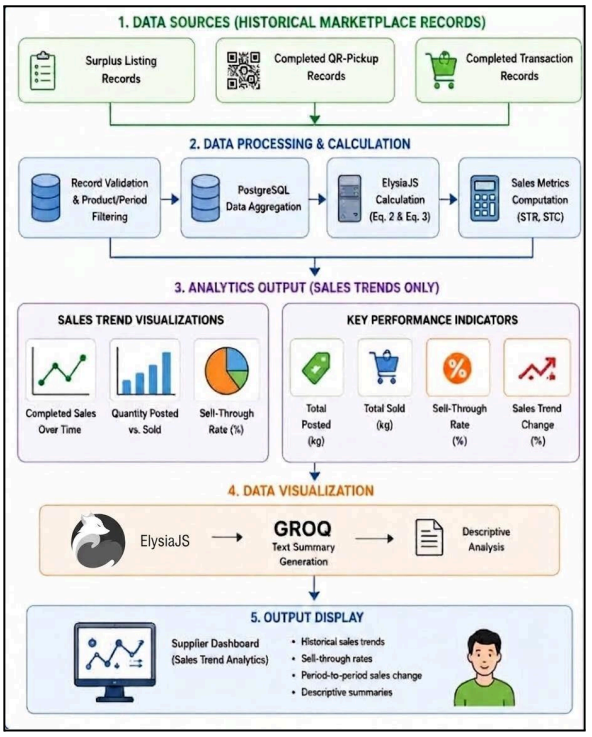

Figure 8. Descriptive Sales Trend Analytics Process

Sales trend analytics uses recorded sales data to describe sales activity across products and periods. Descriptive analytics focuses on analyzing existing data to identify and summarize past performance rather than predict future outcomes [44]. Sales dashboards can also convert tabular transaction records into graphs and summarized measures for easier examination of sales patterns and changes over time [44]. Retail sales visualization systems have been used to present historical sales information through graphical views and trend comparisons [45], [46].

The Descriptive Supplier Sales Trend Analytics Framework organizes Agrivive's completed marketplace records into numerical measures and visual presentations of historical sales activity. In Agrivive, the framework processes surplus listing records, completed QR-validated pickup records, and completed transactions for a selected supplier, product, and period. PostgreSQL stores and aggregates the records, while ElysiaJS performs the required calculations. The results include total quantity posted, total quantity sold, sell-through rate, and changes in completed sales between consecutive periods. The results are then displayed through charts, indicators, and descriptive analysis.

The framework uses the sell-through rate to express the proportion of posted surplus that was recorded as sold through completed pickups. Historical sell-through measures are used in retail analysis to examine how much of available stock has been sold [47]. Agrivive adapts this concept by using completed pickup quantities as the sold quantity and posted surplus quantities as the reference quantity.

The framework also calculates the percentage change in completed sales between consecutive periods. This allows the

system to show whether recorded sales quantity increased or decreased compared with the previous period. Percentage-change calculations are commonly used to compare sales performance across periods [48].

Sell-Through Rate

The framework calculates the sell-through rate for product *p* during period *t* using Equation 2:

$$STR_{p,t} = \frac{Qsold,p,t}{Qposted,p,t} \; x \; 100$$

where:

$STR_{p,t}$ = sell-through of product *p* during period *t*;

$Q_{sold,p,t}$ = total quantity of product *p* recorded as sold through completed QR-based pickups during period *t*; and

$Q_{posted,p,t}$ = total surplus quantity of product *p* posted during period *t*.

Only quantities associated with completed pickup transactions are included in $Qsold_{,p,t.}$ Reserved, cancelled, expired, or returned quantities are not counted as completed sales.

Period-to-Period Sales Trend Rate

The framework also calculates the period-to-period sales trend change using Equation 3:

$$STC_{p,t} = \frac{Qsold,p,t - Qsold,p,t-1}{Qsold,p,t-1} \; x \; 100$$

where:

$STC_{p,t}$ = completed-sale trend change for product p during period t;

$Q_{sold,p,t}$ = completed sales quantity for the current period;

$Q_{sold,p,t-1}$ = completed sales quantity for the immediately preceding period.

A positive result indicates an increase in completed sales, while a negative result indicates a decrease. When the previous period has no completed sales, the system displays the trend rate as not computable rather than performing a division by zero.

Sample Dataset and Computed Results

Table 14. Sample Cabbage Sales Trend Dataset

Period	Quantity Posted	Quantity Sold	Sell-Through Rate	Previous Period Sold	Sales Trend Change	Dashboard Result
Week 1	120 kg	72 kg	60.00%	-	Baseline	Initial Recorded period
Week 2	130 kg	85 kg	65.38 %	72 kg	+18.06 %	Sales increased
Week 3	125 kg	80 kg	64.00 %	85 kg	-5.88%	Sales decreased
Week 4	150 kg	105 kg	70.00 %	80 kg	+31.25 %	Sales increased
Week 5	140 kg	98 kg	70.00 %	105 kg	-6.67%	Sales decreased

The values in Table 11 are prototype data presented only to demonstrate how the framework processes Agrivive records.

Sample Computation for Week 4

During Week 4, the supplier posted 150 kilograms of cabbage, of which 105 kilograms were recorded as sold through completed pickups.

Applying Equation 2:

$$STR_{cabbage,4} = \frac{105}{150} \; x \; 100$$

$$STR_{cabbage,4} = 70\%$$

The cabbage sell-through rate for Week 4 is therefore 70%.

To determine the completed-sales trend from Week 3 to Week 4, Equation 3 is applied:

$$STC_{cabbage,4} = \frac{105-80}{80} \; x \; 100$$

$$STC_{cabbage,4} = 31.25\%$$

The dashboard therefore report:

> *Cabbage completed sales increased by 31.25% from Week 3 to Week 4. A total of 105 kilograms was sold from the 150 kilograms posted, resulting in a 70% sell-through rate.*

This output describes a historical sales trend only and does not forecast buyer demand or predict the quantity of cabbage sold in future periods. PostgreSQL and ElysiaJS perform the quantity aggregation and mathematical calculations. ElysiaJS may organize the summarization process, while Groq converts the validated statistics into a readable explanation. ElysiaJS and Groq do not independently calculate quantities, determine percentages, forecast sales, or generate values outside the validated backend results.

2.3.2.4 Flagging and Trust Mechanism

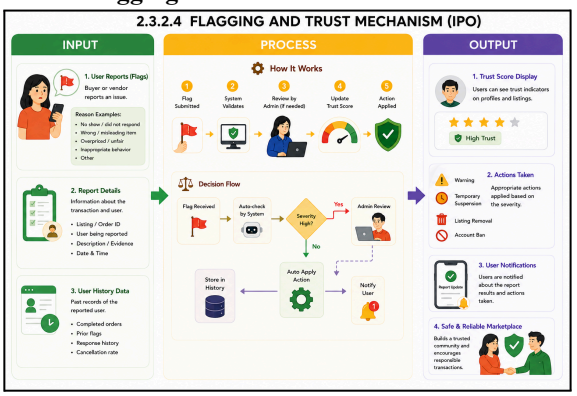

Figure 10. Flagging and Trust Mechanism

The Flagging and Trust Mechanism is a rule-based process for recording and evaluating user activities, transaction feedback, and reported incidents using predefined conditions. Reputation systems in online marketplaces can provide information about user behavior and transaction experiences [54]. Online reputation systems may also be affected by manipulated or fraudulent reviews, which makes the handling of user feedback an important consideration in marketplace systems [54]. Purchase verification can also be used to connect reviews with completed purchases and produce more representative ratings and higher-quality reviews [55].

In Agrivive, buyers can submit ratings, reviews, feedback, and transaction-related reports, which are stored in the database for automated evaluation. The mechanism applies configured rules and a weighted statistical tool to recorded user activity, including inactivity, non-responsiveness, failed reservations, misleading product information, QR-code misuse, inappropriate communication, and other defined violations. Weighted trust models can assign different weights to trust-related factors and combine them into an overall trust score [56]. Weighted evaluation methods can also assign different importance to factors when determining an e-commerce trust evaluation [57]. In Agrivive, these principles are applied to recorded violations by assigning a weight to each violation type and combining the results into a weighted violation score. The resulting score is compared with predefined thresholds to determine the corresponding flagging or account action.

Since vegetables are perishable and online vegetable transactions involve time-sensitive order requirements, Agrivive applies time-based rules to seller responsiveness. Time windows and postharvest quality deterioration are considered in online vegetable retailing, providing a basis for using time-related conditions in vegetable transactions [58]. The specific time thresholds used by Agrivive are system-defined settings.

For seller inactivity or non-responsiveness, Agrivive applies the following rule-based schedule:

Table 17. Flagging and Trust Mechanism Rule

Seller inactivity	System action
12 hours	Send seller a reminder
24 hours	Send a warning and notify the buyer
48 hours	Flag the seller as non-responsive
Repeated violations	Apply temporary restriction or suspension according to the configured rules

The 12-hour, 24-hour, and 48-hour thresholds are Agrivive-defined system rules. They provide specific points at which the system records and responds to seller inactivity.
The mechanism also applies predefined rules to other recorded behaviors. Inactivity for three weeks results in a warning, while one month of inactivity results in account blocking. Non-responsiveness, failure to reply or fulfill a confirmed reservation, and misleading product information may result in a warning followed by stronger account action for repeated violations. These conditions are evaluated from recorded system events rather than from the number of submitted reports alone.

The progressive response is represented through Equation 3, which uses a weighted statistical approach to combine the frequency and assigned weight of recorded violations.

Weighted trust evaluation methods use multiple factors and assign different weights to reflect their contribution to an overall score [56], [57]. Agrivive applies this structure to the recorded violation types.

$$VS_u = \sum_{i=1}^{n} W_i V_{u,i}$$

where:
VS_u = weighted violation score of user u;
w_i = assigned weight of violation type i; and
$v_{u,i}$ = recorded occurrence or value of violation type *i* for user *u*.

For example, a violation type with a higher assigned weight contributes more to the user's total score than a violation type with a lower weight. The assigned weights and score thresholds are Agrivive prototype configurat**ions** and are not taken as fixed values from the cited studies.

The rule-based response can be represented as:

Table 18. Flagging and Trust Mechanism Condition

Rule condition	System action
No violation	No account action
Warning threshold reached	Warning
Repeated violation threshold reached	Temporary restriction
Blocking threshold reached or serious violation	Account blocking

For seller non-responsiveness, the time-based rules operate together with the weighted evaluation. The system first sends reminders and warnings as the inactivity period increases. When the defined threshold is reached, the event is recorded as a non-responsive incident. Repeated incidents contribute to the user's weighted violation score and may result in a stronger account action according to the configured rules.

For example, when a buyer reserves vegetables and does not complete the agreed pickup, the reservation record, QR status, pickup status, and related transaction information are used as inputs for the rule evaluation. When the recorded conditions match a defined violation rule, the incident is included in the user's violation record. An unsupported report does not automatically become a confirmed violation or result in an account penalty.

The mechanism therefore combines time-based rules, transaction records, ratings and feedback, and weighted statistical evaluation. The weighted statistical tool provides the calculation method for combining different violation types, while the rule-based thresholds determine the corresponding system action. The mechanism does not treat every submitted report as a confirmed violation and does not use the weighted score as an independent determination of fraud.

2.3.3 Data Models

2.3.3.1 Use Case Diagram

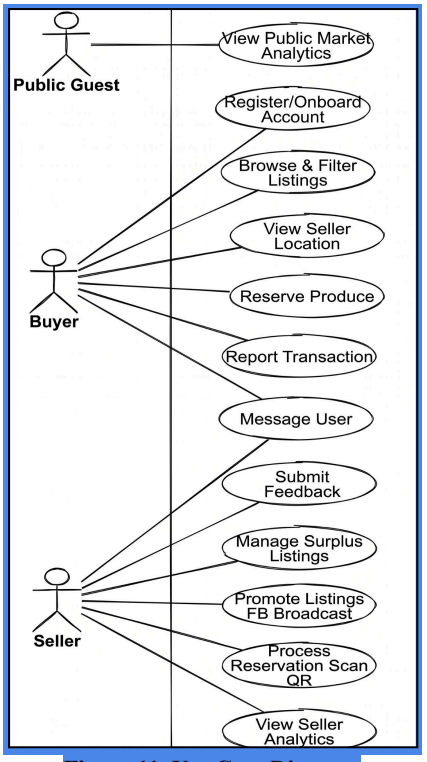

Figure 11. Use Case Diagram

Figure 9 presents the use case diagram of Agrivive, illustrating the interactions among three actors: Public Guest, Buyer, and Seller. Public Guests may view publicly available marketplace analytics and register or onboard into the system. Buyers may browse and filter listings, view seller locations, reserve surplus vegetables, report transactions, message users, and submit feedback. Sellers may manage surplus listings, promote listings through Facebook broadcast, process reservation QR scans, and view seller analytics.

2.3.3.2 Entity Relationship Diagram

An entity-relationship diagram (ERD) visually represents the relationships and associations between entities within a database. It illustrates the overall layout and framework of the database system, highlighting the interactions among different components

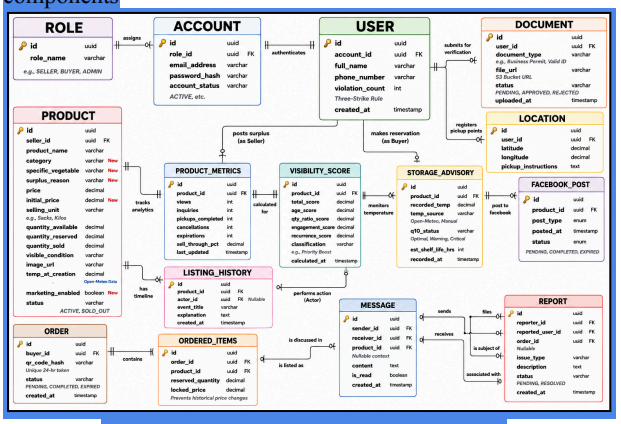

Figure 12. Entity Relationship Diagram

2.3.3.3 Data Dictionary

The following tables are the system's data dictionary.

Table 19. Role

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the role.
role_name	VARCHAR	The specific access level (e.g., 'SELLER', 'BUYER', 'ADMIN').

Table 19 defines the predefined user roles and their corresponding access levels and permissions.

Table 20. Account

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the authentication account.
role_id	UUID	Foreign Key linking to the ROLE table.
email_address	VARCHAR	The user's registered email address, used for login.
password_hash	VARCHAR	The cryptographically hashed password (never stored as plain text).
account_status	VARCHAR	Enum value representing login status ('ACTIVE', 'BLOCKED').

Table 20 supports secure user authentication and access management by separating login credentials from profile information.

Table 21. User

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the user profile.
account_id	UUID	Foreign Key linking to the secure ACCOUNT table.
full_name	VARCHAR	The complete legal name of the user or business representative.
phone_number	VARCHAR	Contact number used for direct communication or logistics.
violation_count	INT	Tracks unresolved complaints used for the Three-Strike Rule.
created_at	TIMESTAMP	The exact date and time the user profile was registered.

Table 21 serves as the primary user record and supports communication, account management, and enforcement of the Three-Strike Rule.

Table 22. Location

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the location record.
user_id	UUID	Foreign Key linking to the specific user.
latitude	DECIMAL	Geographic coordinate used for precise map positioning.
longitude	DECIMAL	Geographic coordinate used for precise map positioning.
pickup_instructions	TEXT	Specific directions for locating the seller's pickup point

Table 22 supports location-based services by helping buyers accurately locate the seller's stall during product pickup using the interactive map.

Table 23. Product

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the surplus listing.
seller_id	UUID	Foreign Key linking to the USER who posted the surplus
product_name	VARCHAR	The name of the product or vegetable being sold.
category	VARCHAR	The general category of the product
specific_vegetable	VARCHAR	The specific type of vegetable being listed.
surplus_reason	VARCHAR	Reason why the product is being sold as surplus.
price	DECIMAL	Seller-defined listing price at the time of publication.
initial_price	DECIMAL	The original price before any adjustments.
selling_unit	VARCHAR	The metric unit of measurement, such as Sacks or Kilos.
quantity_available	DECIMAL	The total amount currently available for reservation.
quantity_reserved	DECIMAL	The total amount currently locked by active QR code reservations.
quantity_sold	DECIMAL	The total amount successfully picked up and paid for.
visible_condition	VARCHAR	Seller-declared condition of the product.
image_url	VARCHAR	URL pointing to the product photograph
temp_at_creation	DECIMAL	Temperature value recorded at the exact moment of posting.
marketing_enabled	BOOLEAN	Indicates whether the listing is enabled for marketing promotion.
status	VARCHAR	Listing visibility state such as ACTIVE or SOLD_OUT.

Table 23 serves as the main inventory record, allowing sellers to manage surplus products while enabling buyers to view accurate and updated product information.

Table 24. Product_Metrics

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the metrics record
product_id	UUID	Foreign Key linking to the PRODUCT table.
views	INT	Total number of times the product listing was viewed.
inquiries	INT	Total number of buyer inquiries regarding the product
pickups_completed	INT	Total number of completed pickups.
cancellations	INT	Total number of cancelled reservations.
expirations	INT	Total number of expired reservations.
sell_through_pct	DECIMAL	Percentage of product sold relative to available stock.
last_updated	TIMESTAMP	Date and time the metrics were last refreshed.

Table 24 supports performance tracking and helps the system evaluate listing engagement, reservation success, and inventory movement.

Table 25. Visibility_Score

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the visibility score record.
product_id	UUID	Foreign Key linking to the PRODUCT table.
total_score	DECIMAL	Overall visibility score computed by the system.
age_score	DECIMAL	Score component based on listing age.
qty_ratio_score	DECIMAL	Score component based on remaining quantity.
engagement_score	DECIMAL	Score component based on user engagement.
recurrence_score	DECIMAL	Score component based on repeated listing behavior.
classification	VARCHAR	Resulting category such as Priority Boost.
calculatet_at	TIMESTAMP	Date and time the score was computed.

Table 25 supports listing prioritization by combining age, quantity, engagement, and recurrence indicators into a single score.

Table 26. Listing_History

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the history record.
product_id	UUID	Foreign Key linking to the PRODUCT table.
actor_id	UUID	Foreign Key linking to the user who performed the action. Nullable.
event_title	VARCHAR	Short title describing the event.
explanation	TEXT	Detailed explanation of the event or listing change.
created_at	TIMESTAMP	Date and time the history event was recorded.

Table 26 provides an audit trail of changes such as updates, boosts, and status modifications.

Table 27. Storage_Advisory

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the advisory record
product_id	UUID	Foreign Key linking to the PRODUCT table.
recorded_temp	DECIMAL	Temperature recorded for storage analysis.
temp_source	VARCHAR	Source of temperature data, such as Open-Meteo or Manual.
q10_status	VARCHAR	Storage condition status such as Optimal, Warning, or Critical.
est_shelf_life_hrs	IINT	Estimated shelf life in hours.

recorded_at	TIMESTAMP	Date and time the advisory data was recorded.

Table 27 supports inventory preservation by helping the system estimate freshness and shelf life.

Table 28. Facebook_Post

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the Facebook post record
product_id	UUID	Foreign Key linking to the PRODUCT being promoted.
post_type	ENUM	Type of post, such as FIRST_POST or SECOND_POST.
posted_at	TIMESTAMP	Date and time the product was posted on Facebook
status	ENUM	Status of the post, such as SUCCESS or FAILED.

Table 28 supports the platform's marketing and social broadcasting features by tracking posted products, post sequence, and publication success or failure.

Table 29 . Order

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the reservation ticket.
buyer_id	UUID	Foreign Key linking to the USER making the reservation.
qr_code_hash	VARCHAR	A unique encrypted string used to generate the 24-hour QR code
status	VARCHAR	Transaction lifecycle state such as PENDING, COMPLETED, or EXPIRED.
created_at	TIMESTAMP	Time the reservation was created

Table 29 supports the reservation process by tracking transactions and ensuring that reserved products are claimed within the allowed reservation period.

Table 30. Ordered_Items

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the line item.
order_id	UUID	Foreign Key linking to the parent ORDER
product_id	UUID	Foreign Key linking to the specific PRODUCT reserved.
reserved_quantity	DECIMAL	The exact amount of the product requested by the buyer.
locked_price	DECIMAL	Stores the price at the time of reservation to prevent changes if the seller updates the live price later.

Table 30 supports order management by allowing multiple products to be included in a single reservation while preserving the original transaction details.

Table 31. Message

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the chat message.
sender_id	UUID	Foreign Key linking to the USER sending the message.
receiver_id	UUID	Foreign Key linking to the USER receiving the message.
product_id	UUID	Nullable foreign key providing context if the message is related to a specific listing.
content	TEXT	The actual written text of the communication.
is_read	BOOLEAN	Indicates whether the receiver has opened the message.
created_at	TIMESTAMP	Date and time the message was sent.

Table 31 supports communication within the platform by allowing users to discuss product inquiries and reservation details directly.

Table 32. Report

Field Name	Data Type	Description
id	UUID	Primary Key. Unique identifier for the filed report.
reporter_id	UUID	Foreign Key linking to the USER submitting the complaint.
reporter_user_id	UUID	Foreign Key linking to the USER accused of the violation
order_id	UUID	Nullable foreign key linking the report to a specific transaction.
issue_type	VARCHAR	The category of complaint, such as Ghosting, Bogus Seller, or Condition Mismatch.
description	TEXT	The detailed explanation provided by the reporter.
status	VARCHAR	Administrator resolution state such as PENDING, RESOLVED, or DISMISSED.
created_at	TIMESTAMP	Date and time the report was filed.

Table 32 supports system accountability by allowing administrators to review reported issues, resolve disputes, and enforce the Three-Strike Rule for repeated policy violations.

2.3.3.4 Architecture Design

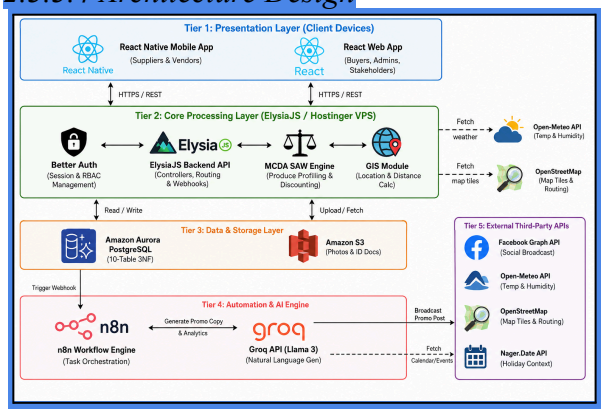

Figure 13. Architecture Design

This diagram shows the system architecture, providing a quick overview of the platform's technical structure. It displays a five-tier layout including the client devices, core processing layer, data and storage layer, automation and AI engine, and external third-party APIs. Also, they display the connections between the mobile app, web app, backend systems, databases, and external tools like weather and maps. This diagram is an efficient method for visualizing how the entire digital system operates and connects.

2.3.3.5 Sample Prototype

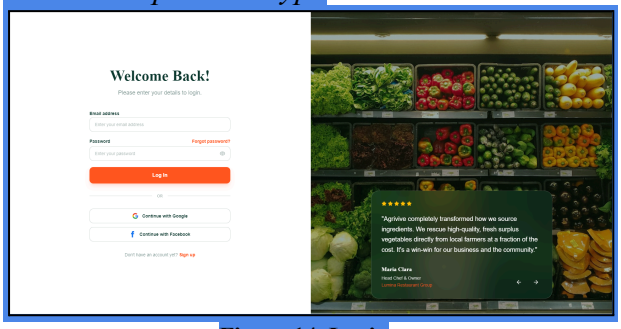

Figure 14. Login

This UI displays the **Login section**, which includes a welcome message, fields for entering an email address and password, a login button, and options to continue with Google or Facebook for returning users.

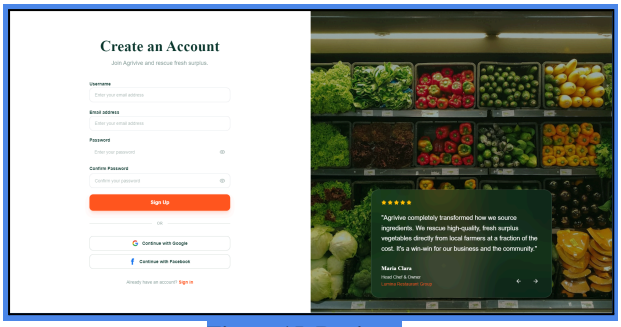

Figure 15. Register

This UI shows the **Register section** for creating a new account. It includes fields for entering a username, email address, password, and password confirmation, along with a sign-up button and options to register using Google or Facebook.

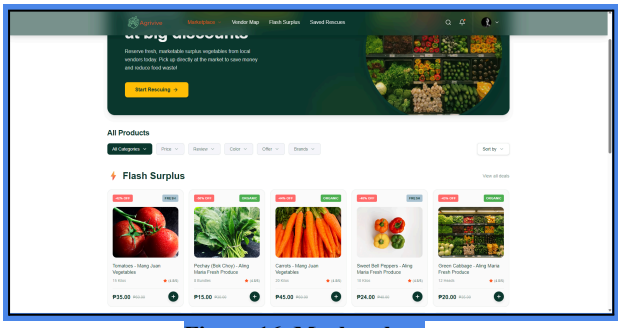

Figure 16. Marketplace

This UI displays the **Marketplace section**, where buyers can browse available surplus vegetables and filter products by category, price, reviews, and brand. Each product displays its image, seller name, product weight, discount percentage, original price, discounted price, and a reservation button.

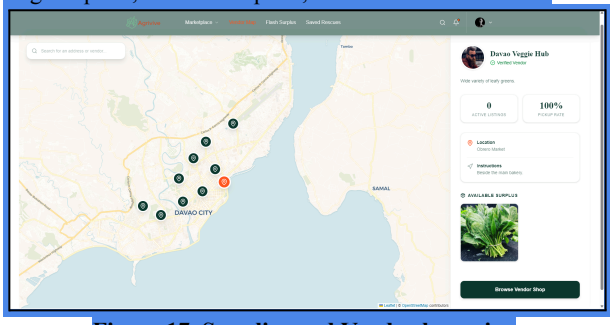

Figure 17. Supplier and Vendor browsing

This UI displays the **Supplier and Vendor Browsing section**, featuring an interactive map on the left that shows the physical locations of market stalls using pin icons.

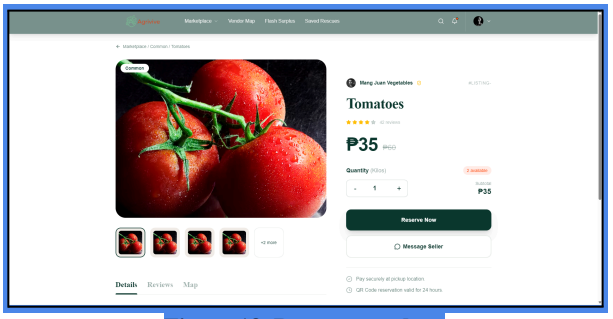

Figure 18. Reserve produce

This UI displays the **Product Reservation page**, which shows the selected product image, price, and quantity selection. Users can adjust the quantity using the add and remove buttons and reserve the product by clicking the reservation button.

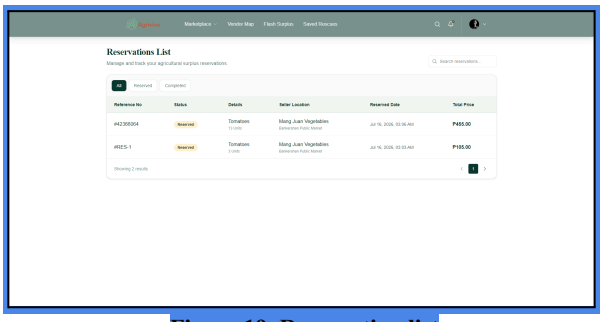

Figure 19. Reservation list

This UI displays the **Reservation List section**, which shows all reserved products in a table for easy monitoring. The table includes the reservation number, reservation status, product name, seller location, reservation date and time, and total amount to be paid.

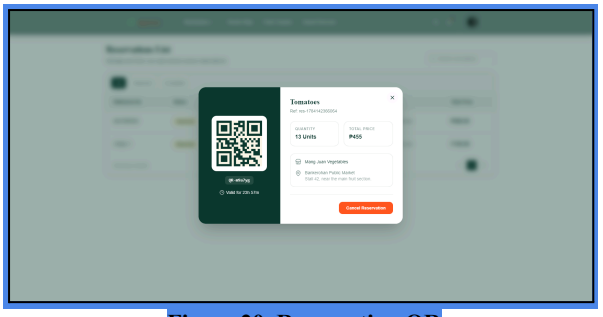

Figure 20. Reservation QR

This UI displays the **Reservation Confirmation** window after a successful reservation. It generates a QR code for product pickup. It also displays the reserved quantity, total payment, pickup instructions, and a cancel reservation button.

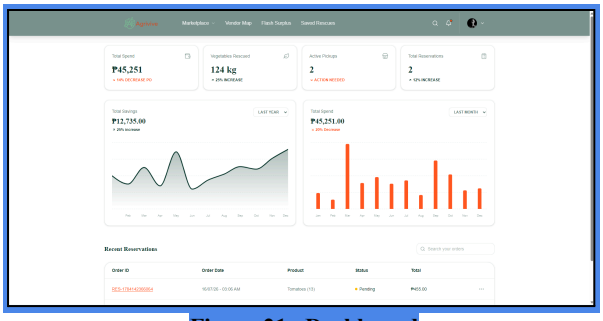

Figure 21. Dashboard

This UI displays the **Dashboard section**. It provides a summary of the user's activities. It displays the total amount spent, total vegetables rescued, active reservations, completed orders, savings over time, spending trends, and the list of recent transactions.

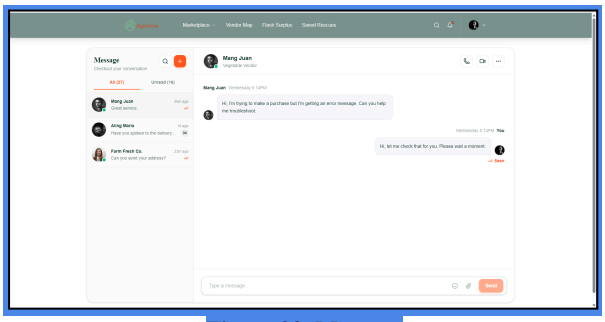

Figure 22. Message

This UI displays the **Messaging section**. Where users can communicate directly with vendors through the built-in chat feature. It displays the conversation list, chat history, and a message box where users can send messages regarding reservations and product inquiries.

2.4 Application Development

The application development of Agrivive followed an iterative Scrum agile framework across six planned sprints during Academic Year 2026–2027. Development targeted a dual-stack cross-platform ecosystem: a Next.JS web application for buyers, administrators, and public stakeholders, and a React Native mobile application for verified vegetable suppliers and market vendors. The backend processing layer was constructed using ElysiaJS and Better Auth, with PostgreSQL managing core relational entity storage, OpenStreetMap/React Leaflet powering stall geolocation, n8n handling workflow automation, and Groq supporting natural language promotional drafting and analytical summaries.

Table 28 outlines the sprint breakdown, functional mapping, and core technical deliverables completed across the development lifecycle.

Table 28. Sprint Summary for Agrivive Functional Requirements

Sprint	Module Focus	Target Functional Requirements	Core Deliverables & Outputs
Sprint 1	Core Architecture & Authentication	FR-01, FR-02, FR-16	ElysiaJS API setup, Better Auth OTP verification, PostgreSQL schema initial deployment, Next.JS landing & web auth, React Native mobile login layout, public stakeholder dashboard base.
Sprint 2	Seller Listing & Inventory Management	FR-03, FR-05	Mobile product creation form (quantities, pricing, selling unit), declared condition tracking, image uploads via Amazon S3, stock tracking (available/reserved/sold), visibility tier display.
Sprint 3	Buyer Discovery & Stall Geolocation	FR-10, FR-14	Next.JS marketplace catalog, multi-criteria filtering (category, price, location), OpenStreetMap & React Leaflet interactive stall map, pickup instructions display.
Sprint 4	Reservation, QR Scanning & Pickup Handover	FR-06, FR-12, FR-13	24-hour stock reservation engine, React QR generation for buyers, React Native mobile QR scanner for sellers, automatic stock restoration upon cancellation/expiration, transactional handover lock.
Sprint 5	Marketing Boost & User Communication	FR-04, FR-05, FR-11	AHP-TOPSIS priority trigger to n8n webhooks, Groq (GPT-OSS-20B) promo copy generation, 6-hour auto-discounting workflow, cross-platform in-app messaging, push/buyer notification system.
Sprint 6	Supplier Analytics, Advisories & Trust Mechanism	FR-07, FR-08, FR-15	Descriptive sales trend analytics dashboard (STR & STC calculations), Nager.Date holiday advisories integration, weighted statistical violation tool, transaction rating/review/flagging engine.

2.4.1 Sprint 1 – Core Architecture, Authentication & Stakeholder Portal

Sprint 1 established the foundational infrastructure of the Agrivive platform. The development team configured the ElysiaJS backend API running on Node/Bun inside Docker containers on Hostinger VPS, connected to a PostgreSQL database. Better Auth was implemented to govern secure user registration, email verification via One-Time Passwords

(OTP), and Role-Based Access Control (RBAC) across three distinct user types: Sellers, Buyers, and Stakeholders/Admins. The initial responsive Next.JS web layout was deployed alongside the primary React Native mobile shell, establishing base endpoints for public stakeholder data access without exposing personally identifiable information.

2.4.2 Sprint 2 – Mobile Surplus Listing & Inventory Management

Sprint 2 focused on empowering verified vegetable suppliers and retail vendors to publish and track marketable surplus. The React Native mobile app was enhanced with structured listing creation workflows, allowing sellers to define vegetable categories, specific produce types, total quantities, custom selling units (e.g., sacks, kilograms), initial pricing, minimum price floors, and declared storage conditions. Integration with Amazon S3 was finalized for uploading recent product photographs. The backend schema in PostgreSQL was expanded to maintain real-time inventory balances, separating posted stock into available, reserved, and sold quantities while rendering active listing visibility statuses to the seller interface.

2.4.3 Sprint 3 – Buyer Marketplace Discovery & Interactive Stall Geolocation

Sprint 3 addressed buyer product discovery and vendor location transparency. The Next.JS web application catalog was developed, enabling household and business buyers to search surplus listings using multi-criteria filters based on produce category, price range, seller rating, and geographic proximity. Map-based stall localization was integrated into the frontend using React Leaflet and OpenStreetMap tile services, plotting verified seller pins across Bankerohan Public Market, Agdao Public Market, and the Davao Food Terminal Complex. Buyers were provided with precise location details, distance calculations, and seller-provided pickup instructions prior to committing to a transaction.

2.4.4 Sprint 4 – 24-Hour Stock Reservation & Mobile QR Pickup Verification

Sprint 4 established the transaction and reservation engine. Logic was implemented in ElysiaJS to allow buyers to lock available surplus quantities into a 24-hour reservation window. Upon confirmation, the Next.JS web app utilizes React QR Code to generate a unique, time-stamped QR reservation ticket. On the seller side, the React Native mobile app was integrated with a QR scanner library, allowing vendors to scan buyer QR tickets at physical market stalls, validate token authenticity, complete product handover, and log cash-on-pickup sales. Background worker tasks were configured in PostgreSQL to automatically restore locked quantities to available stock if a reservation expires uncollected or is cancelled by the buyer.

2.4.5 Sprint 5 – Automated Marketing Boost & Cross-Platform Communication

Sprint 5 integrated automated promotional workflows and real-time user communication. The AHP-TOPSIS decision engine was tied to n8n webhook triggers; listings qualifying for "Priority Boost" automatically dispatch payload data to n8n. The workflow sends sanitized details to Groq (running GPT-OSS-20B) to generate engaging marketing captions and shareable listing links, storing drafts in PostgreSQL. A 6-hour follow-up scheduler was implemented in ElysiaJS to apply seller-configured price reductions (respecting price floors) and trigger updated promotional content. In-app messaging and notification modules were built into both frontend client stacks to enable direct buyer-seller inquiry coordination.

2.4.6 Sprint 6 – Supplier Sales Trend Analytics, Advisories & Trust Mechanism

Sprint 6 finalized the analytical and trust-enforcement components. The Descriptive Supplier Sales Trend Analytics module was deployed, calculating Sell-Through Rates (STR) and Period-to-Period Sales Trend Changes (STC) from completed QR pickup records, with Groq rendering text-based analytical summaries on seller dashboards. Calendar advisories were integrated via the Nager.Date API to supply Philippine public holiday context. Finally, the Flagging and Trust Mechanism was built, implementing time-based inactivity warnings (12h/24h/48h) and a weighted statistical violation score model to process buyer reviews, ratings, and fraud reports for account accountability.

2.5 Testing

To ensure system reliability, performance, security, and usability across both web and mobile platforms, comprehensive testing was conducted on the Agrivive system. The evaluation encompassed functional testing, non-functional testing, data visualization, security, workflow automation, and recommendation engine scenarios, followed by alpha and beta evaluations with target users in Davao City.

2.5.1 Functional Testing

Functional testing verified that all system modules perform according to the specified requirements. Test cases were executed across buyer, seller, and administrative user roles.

Table 29. Functional Test Cases for Buyer Functionalities

Test ID	Test Scenario	Expected Output	Actual Result	Status
FT-B-01	Account registration with email OTP verification	OTP delivered; account created upon valid input	Account verified and logged in	Pass
FT-B-02	Multi-criteria surplus product search and filtering	Listings filtered accurately by category, price, and location	Listings matched selected criteria	Pass
FT-B-03	24-Hour QR Reservation generation	QR code generated; available stock reserved for 24h	QR code rendered and stock updated	Pass

FT-B-04	Interactive stall map viewing (OpenStreet Map/Leaflet)	Vendor stall pin and pickup instructions displayed	Map pin rendered correctly with details	Pass
FT-B-05	Submit rating, review, and fraud report	Feedback logged in database and reflected on profile	Review stored and rating re-calculated	Pass

Table 30. Functional Test Cases for Seller Functionalities

Test ID	Test Scenario	Expected Output	Actual Result	Status
FT-S-01	Surplus vegetable listing creation with photos	Listing saved, photo uploaded to S3, status active	Listing created with valid S3 image URL	Pass
FT-S-02	Mobile QR scanner verification during pickup	QR validated, reservation completed, quantity_sold updated	Reservation marked completed successfully	Pass
FT-S-03	View visibility tier and promotional captions	Displays calculated tier and Groq AI generated copy	Visibility tier and captions displayed	Pass
FT-S-04	Descriptive sales trend analytics dashboard viewing	Dashboard displays STR, STC, and text summary	Metrics calculated and rendered correctly	Pass

2.5.2 Non-Functional Testing

Non-functional testing evaluated browser compatibility, device screen responsiveness, and system performance under varying load conditions.

Table 31. Browser and Device Compatibility (Web & Mobile Applications)

Platform / Browser	Version Tested	Observations	Status
Google Chrome	v122.0	Full layout and interactive map compatibility	Pass
Mozilla Firefox	v123.0	Seamless rendering and WebSocket messaging	Pass
Microsoft Edge	v122.0	All features functional without alignment issues	Pass
Android OS (Mobile)	Android 10 - 14	Camera scanning and push notifications operational	Pass

Table 32. Screen Resolution Compatibility

Device Type	Resolution / Width	Layout Responsiveness	Status
Mobile Phone	360px × 800px	Mobile stacked layout rendered clean	Pass
Tablet	768px × 1024px	Grid re-aligned automatically	Pass
Desktop Display	1366px × 768px+	Full multi-column web layout active	Pass

Table 33. Performance Testing Results

Action / Endpoint	Benchmark Threshold	Average Response Time	Assessment
Catalog Search & Filter Query	< 2.0s	0.42s	Optimal
QR Code Verification Scan	< 1.5s	0.28s	Optimal
AHP-TOPSIS Score Calculation	< 3.0s	0.65s	Optimal
Analytics Summary Generation	< 4.0s	1.85s	Optimal

2.5.3 Data Visualization & Analytics Testing

Testing verified that analytics calculations (STR and STC) produce mathematically sound figures and that chart visualizations render data accurately without misrepresentation.

Table 34. Data Visualization Testing

Component	Test Condition	Expected Outcome	Status

Sell-Through Rate (STR)	Calculated over completed pickups	STR % matches (Qsold / Qposted) * 100	Pass
Sales Trend Change (STC)	Period-to-period sales delta	STC % computed; 'not computable' if baseline is 0	Pass
Dashboard Visuals	Dynamic data rendering	Charts accurately plot weekly volume trends	Pass

2.5.4 Data Security & Authentication Testing

Security testing evaluated authentication controls, access restrictions, session management, and encrypted data transmission across endpoints.

Table 35. Data Security & Authentication Testing

Security Check	Test Execution	Observed Result	Status
Password Hashing	Database record inspection	Passwords stored exclusively as cryptographic hashes	Pass
Role-Based Access (RBAC)	Attempt unauthorized API request	Unauthorized access blocked with HTTP 403 Forbidden	Pass
Data Transmission Security	Inspect traffic during login and API calls	All traffic encrypted using HTTPS/TLS	Pass
OTP Verification	Input expired/invalid OTP	Access rejected; request for new OTP prompted	Pass

2.5.5 Recommendation Model & Workflow Test Scenarios

Specialized test scenarios were designed to evaluate the core algorithmic and automation features of Agrivive, including the AHP-TOPSIS Visibility Engine, n8n Marketing Boost Mechanism, Descriptive Analytics Framework, and Flagging and Trust Mechanism.

Table 36. Algorithmic and Workflow Test Scenarios

Mechanism	Test Scenario	Expected System Behavior	Status
AHP-TOPSIS Engine	Listing evaluation with score >= 0.60	Assigned 'Priority Boost' tier; triggers n8n marketing webhook	Pass
n8n Marketing Boost	Priority Boost webhook trigger & 6h auto-discount	Groq generates promo copy; ElysiaJS updates price after 6h	Pass
Descriptive Analytics	Compute sell-through rates & Groq summary	ElysiaJS calculates STR/STC; Groq formats readable text	Pass
Flagging & Trust	48h	System issues reminder at 12h, warning at 24h, flags non-responsive at 48h	Pass

2.5.6 User Acceptance & Beta Testing

Field evaluations were conducted with produce sellers and buyers from Bankerohan Public Market, Agdao Public Market, and the Davao Food Terminal Complex. Feedback confirmed high usability and operational effectiveness across all evaluated features.

2.5.1 Functional Testing

Table 29. Test Cases for User Functional Testing
Table 30.

2.5.2 Non-Functional Testing

Table 31. Browser and Device Compatibility (Web Application)
Table 32. Screen Resolution Compatibility
Table 33. Performance Testing Results

2.5.3 Data Visualization Testing

Table 34. Data Visualization Testing

2.5.4 Data Security & Authentication Testing

Table 35. Data Security & Authentication Testing

2.5.5 Recommendation Model-Test Scenario

2.5.6 User Acceptance Testing

Figure 34. User Acceptance Checklist (1)
Figure 35. User Acceptance Checklist (2)
Figure 36. User Acceptance Checklist (3)

2.6.7 Alpha Testing

Figure 37. Alpha Testing Form
Figure 38. Proof of Activity (1)

2.6.8 Beta Testing

Table 40. Beta Tester Average Scores
Figure 39. Proof of Activity (2)
Figure 40. Proof of Activity (3)

Figure 41. Proof of Activity (4)
Figure 42. Proof of Activity (5)
2.5.9 Implementation
2.5.9.1 Installation
2.5.9.2 Network Configuration
2.5.9.3 Database Setup
2.5.9.4 User's Manual
2.5.9.5 User Training
Figure 43. User Training
Figure 44. Admin User Acceptance Testing Checklist
Figure 45. Admin User Acceptance Testing Checklist
Figure 46. Certificate of Deployment
Figure 47. Certificate of Acceptance
Figure 48. Certificate of Presentation
3. RESULTS AND DISCUSSION
4. CONCLUSION AND RECOMMENDATION
4.1 Conclusion
4.2 Recommendation
4.2.1 User Recommendation
4.2.2 Researcher's Recommendation
4.2.3 Public Defense Expert Recommendation

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APPENDIX G

Other Figures, Forms, and Table

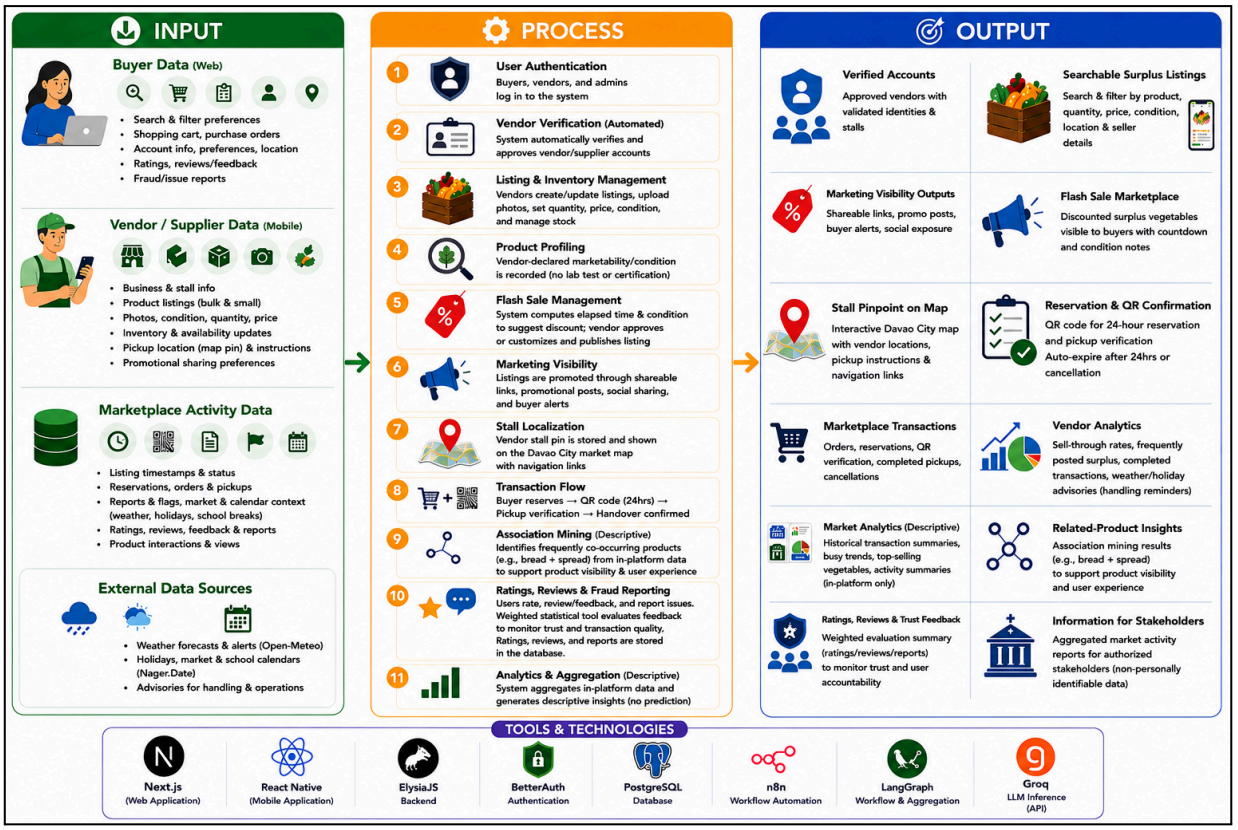

Figure 3. Conceptual Framework

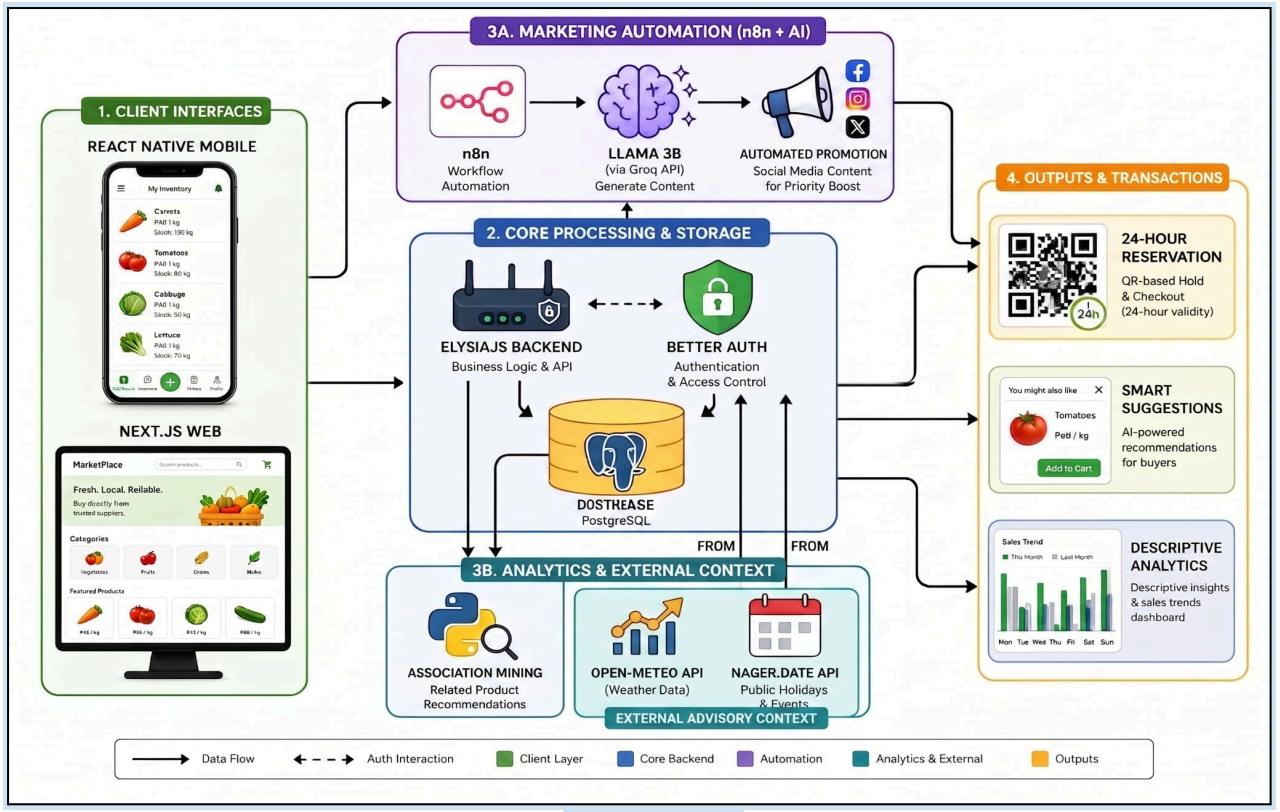

Figure 4. Pipeline

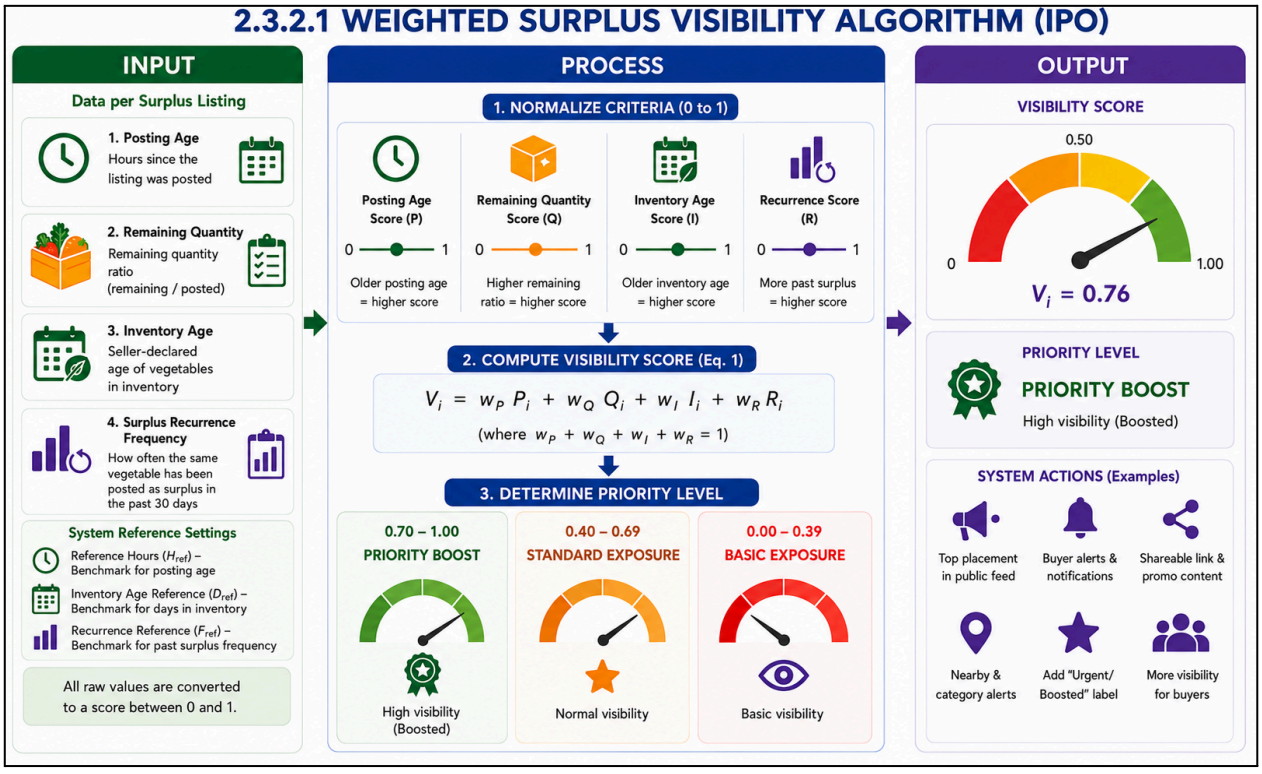

Figure 5. Weighted Surplus Visibility Algorithm

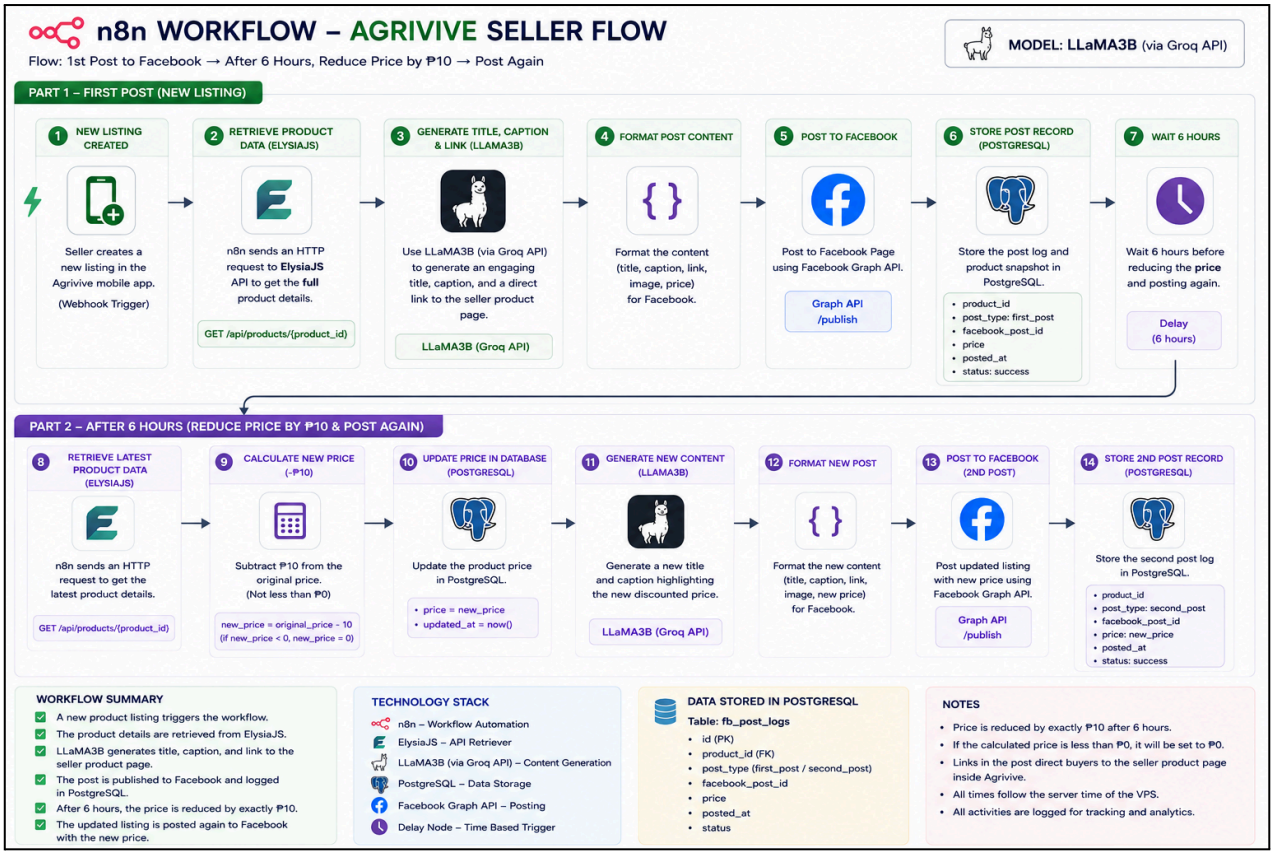

Figure 6. n8n Marketing Boosting Mechanism

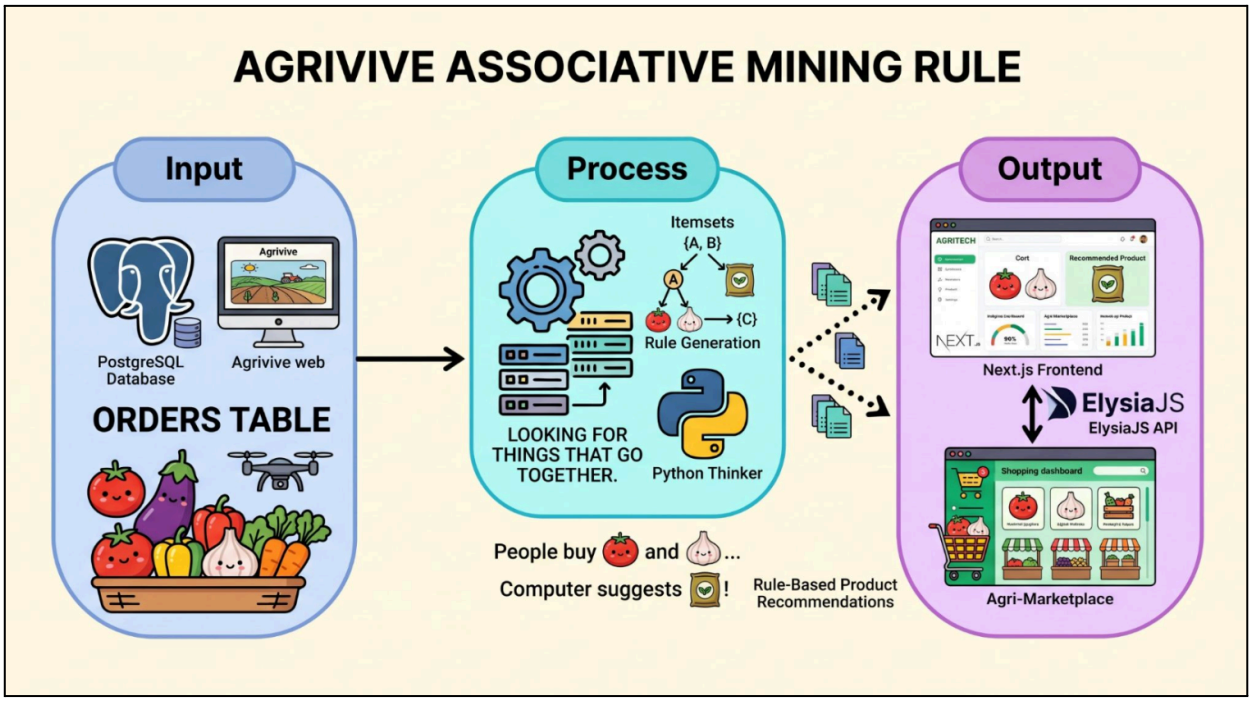

Figure 7. Market Basket Analysis for Related-Product Recommendations

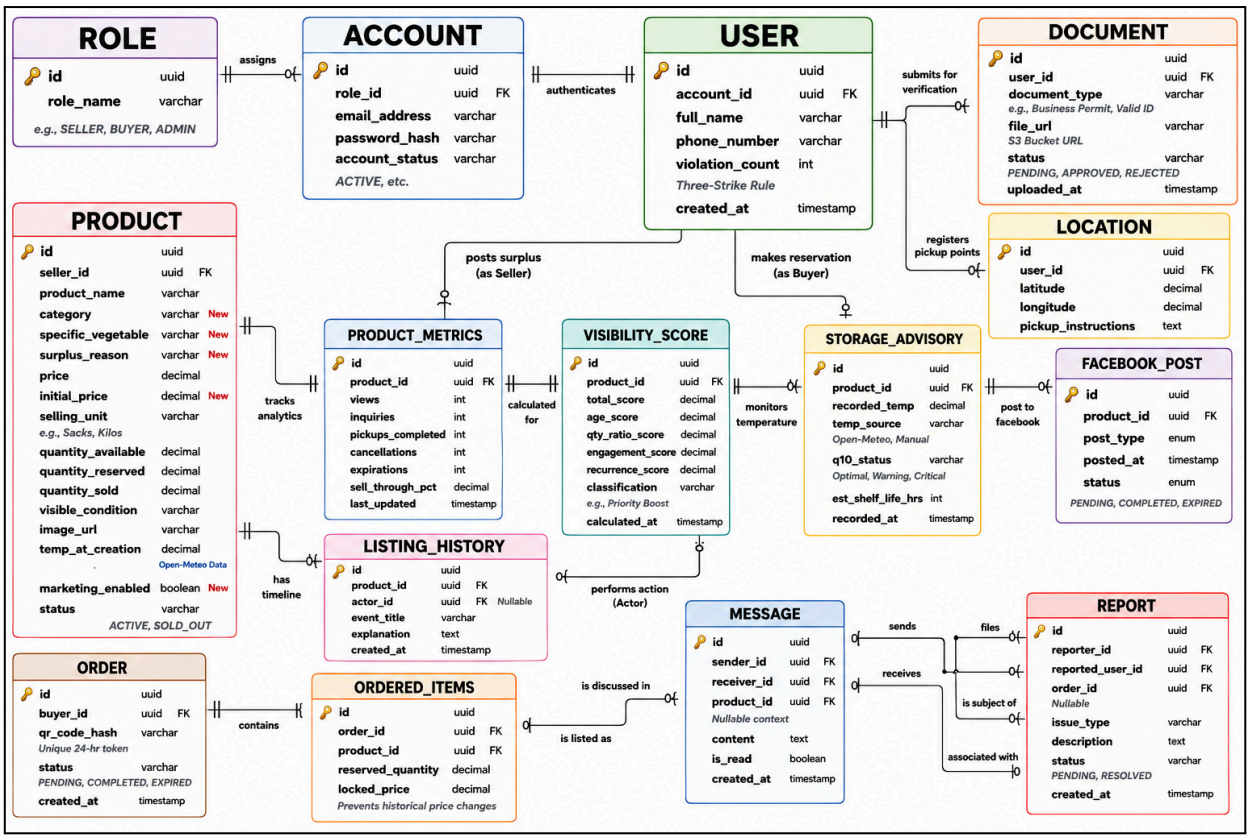

Figure 12. Entity Relationship Diagram

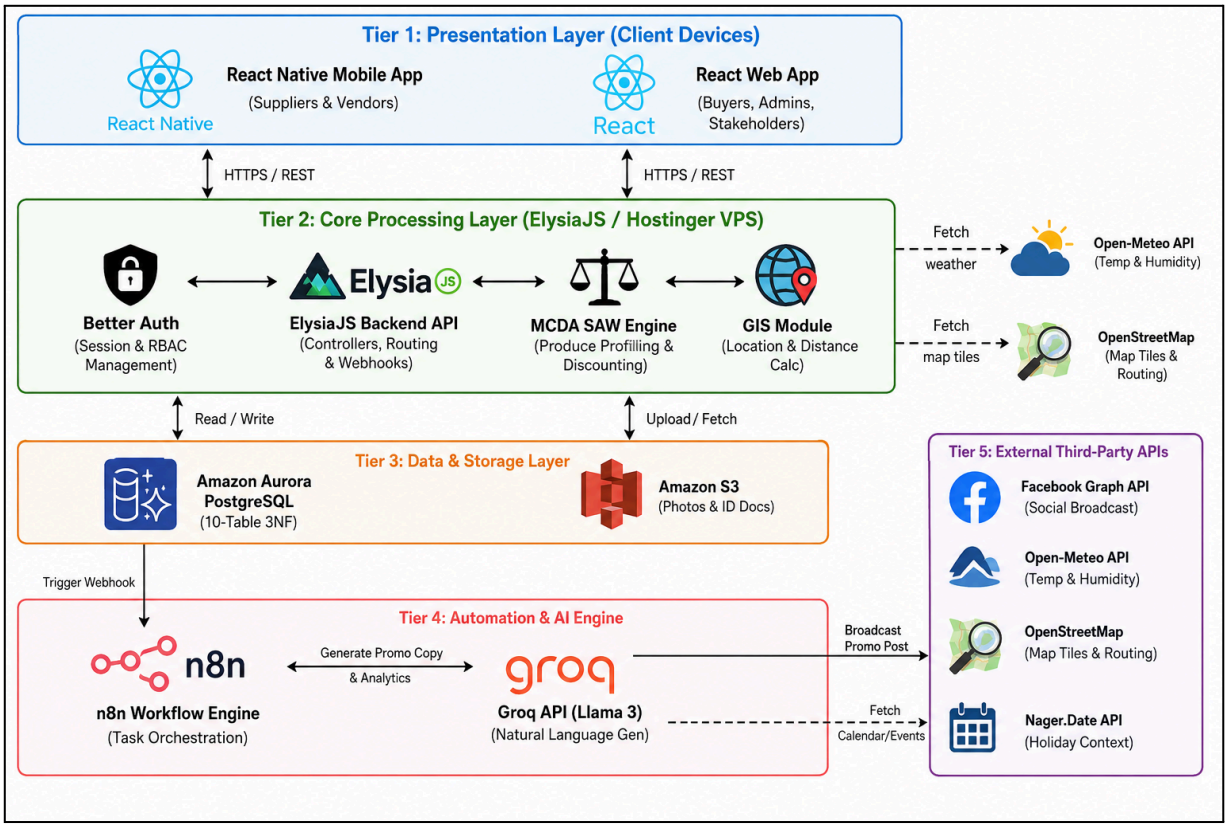

Figure 13. Architecture Design